

New Exporter Dynamics of Born-Global and Multinational Firms*

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Abstract

Several international business studies have found that the internationalization of firms is a gradual process, with successful new exporters contributing to a large share of medium- and long-term export growth. In this paper we examine whether the dynamics of new exporters differ when they are multinational companies (MNCs) and 'born-global' firms (BGs). Previous work finds that MNCs and BGs account for a large share of exports by new exporters, and suggests that their entry, growth and exit behavior resembles that of experienced exporters (Blum et al., 2020). To analyze the export strategy of MNCs and BGs, we exploit disaggregated and comprehensive data on French exporters between 1993 and 2006. We find that MNC and BG new exporters, like other new exporters, start small, grow fast, and exit more at the beginning of each firm-product-destination spell. In addition, age dependence is more pronounced in initial spells, and successful new exporters are more likely to expand early into new markets, suggesting dynamic complementarities in export entry decisions.

JEL codes: F14, F23

Keywords: Firm Dynamics; Exports; Multinational Firms; Born-Global Firms

1 Introduction

Aggregate trade is concentrated in the hands of a few 'superstar exporters' (Bernard et al., 2018, p.567), shaping the export patterns of (also developing) countries (Freund and Pierola, 2015). Freund and Pierola (2020) find that for 32 developing countries, the top five exporters are primarily foreign-owned, account for nearly a third of exports, and about half of export growth. Despite not knowing much about how do 'superstar exporters' become such, new exporters have

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been shown to grow surprisingly fast (i.e. they can account for about half of total exports over a 10 year period even if their initial contribution is small, see Eaton et al. (2008); Albornoz et al. (2023)). Two types of new exporters seem key to understanding how new exporters grow to gain macroeconomic relevance: multinational corporations (MNCs) and firms that export from year 1 (Born-Global firms or BGs). Blum et al. (2020) provide evidence of BGs' prominence in the contribution of new exporters to total export growth. For instance, almost 85 percent of Chile's export growth due to new exporters over 5-year periods comes from BGs that succeed in exporting for at least 4 years. In addition, Inagaki (2025) shows that how BGs grow is crucial for the responsiveness of exports to changes in trade policy. Yet, not much is known about how do either MNCs or BGs grow.

In this paper, we empirically examine whether MNC and BG new exporters behave like experienced exporters or whether their behavior more closely resembles that of other, smaller new exporters. We know from previous research that these two types of firms are different. MNCs have significantly different characteristics from non-multinationals, in particular intangible assets (Antràs and Yeaple, 2014). BGs also own superior internal resources and capabilities, including physical, human and intangible assets, such as labor, capital or organizational and technological skills (Knight and Cavusgil, 2004). The question is whether they differ from other smaller exporters from inception, or whether they grow and expand differently, to become who they are?

Here, we develop a method to identify age dependence in export performance in entry, exit and growth among new exporters. If we identify similar age effects when contrasting MNCs (BGs) with non-MNC (BG) firms (respectively), then we will conclude that bigger (MNC-BG) and smaller new exporters grow similarly. To do so, we exploit highly disaggregated data on French exporters in the 1993-2006 period, from merging three different sources of data: customs data on the universe of exports (by manufacturing firm, country, HS6 product and year); data on foreign affiliates of French multinationals and foreign-owned French firms (LIFI); and balance sheet data on all firms incorporated in France (FICUS).

First, we explore the dynamics of firm-country-product export spells. In particular, we look for spell age effects in export growth and exit probability within a spell, controlling for firm size as well as other firm, country and product characteristics, such as firm TFP growth.¹ We fit a

¹Shocks to firm TFP growth are the main determinant of foreign market entry in the canonical model of

linear model of Foreign sales export growth rates and a non-linear model of spell exit hazard using a nonparametric specification for age. We find that both export growth rates and exit hazards fall with spell age for all new exporters, including Multinational and Born-Global firms. In the case of growth regressions, MNCs and BGs display the same age pattern as other smaller new exporters. But when it comes to exit, MNCs and BGs have a lower exit probability at any age than smaller, new exporters, even if hazard regressions display a similar decreasing in age qualitative pattern for both groups of new exporters.

Second, we also examine potential dynamic complementarities across country-product spells within an exporting firm. Comparison of initial growth/exit/entry in first-ever spells with subsequent spells identifies differential effects by years of experience as an exporter across markets and/or products. We claim that higher age dependence in a first-ever spell relative to subsequent spells, controlling for firm size and productivity growth, is consistent with a gradual internationalization process stemming from current market access being informative about future entries' success probabilities, i.e. from dynamic complementarities in time across countries and products.² Focusing on MNC and BG exporters, we test whether newly exporting firms experience higher growth, more entry and more exit early in their first-ever spell relative to later spells. We find that export growth rate differentials in the first year of the first country-product spell are the same for MNCs and BGs as for other firms. Conditional on surviving, both MNC and BG new exporters are more likely to enter with another product in the same country or with the same product in another country, immediately after their first entry. This pattern also holds among non-MNC/non-BG new exporters, albeit slightly more when adding products and slightly less so when adding countries. Finally, we find that immediate exit is more common in first spells than in later spells among new exporter BGs, even though immediate exit is less frequent overall compared to non-BG new exporters. However, MNCs are not significantly more likely to exit early in their first spell relative to subsequent spells. To pool single- and multi-spell exporters, we avoid using firm fixed effects, so the results might reflect unobserved composition effects among exporters. The overall picture, however, is that MNCs and BGs display qualitatively similar age-dependence as smaller new exporters do, particularly early in their exporting

new exporter dynamics. See Alessandria et al. (2021b) for a survey of recent novel results and open questions remaining within the canonical model of new exporter dynamics.

²See Albornoz et al. (2023) for a similar methodology but applied to the sub-sample of multi-product exporters in the same dataset. There, as here, dynamic complementarities across product-market spells and periods could come from correlations in firm profitability across spells that persist in time, even if entry costs are similar and firm profitability is unknown but can be uncovered upon entry.

life-cycles. We are thus uncovering novel evidence suggesting that French MNCs and BGs face significant uncertainty about their performance in foreign markets, or significant frictions in reaching their optimal foreign sales, despite their substantial international experience. They typically enter small, but quickly expand their product scopes and geographical presence upon surviving, becoming more profitable in their new export spells.

Our work is related to several strands of the literature on firms in international trade. Firstly, this paper relates to the literature on individual export dynamics recently surveyed in Alessandria et al. (2021a), and in particular, to research within it showing evidence of age dependence³. Age dependence refers to new exporters’ behavior that changes as firms become more experienced, controlling for size. Typical age patterns identified are higher exit probabilities right after entry, or higher early foreign sales growth than later in the exporter’s life cycle. It is also directly related to Blum et al. (2020), who find that newly exporting Chilean firms are often owned by multinationals and/or born global, possibly as a result of a spin-off from experienced large exporters. These “atypical” new exporters start as relatively large exporters, are significantly less likely to quit exporting in the first year, and export a stable high share of their sales. These findings speak to a literature exploring long-term size and growth differences between BGs and other firms, some of which examines internationalization. For example, using data on Danish manufacturing start-ups, Choquette et al. (2017) show that BGs display higher turnover, employment, job growth, and a wider geographic reach than comparable firms that internationalize later, but little or no productivity advantage. For France, Bellone et al. (2023) find that BGs display broader export scope rather than higher average export quality, with the clearest ex post gains arising from (received) support to market expansion. Taken together, BGs appear indeed different because of a distinct combination of early export intensity, broader market reach, and post-entry dynamic expansion, leaving open the question of whether their product-market (or within-spell) growth and survival patterns fundamentally differ from those of other new exporters.

Secondly, we also relate to work on the internationalization process of MNCs. Pioneering work in management science by Johanson and Vahlne (1977) offered a model of the internationalization

³See, among others, Eaton et al. (2008), Freund and Pierola (2010), Alborno et al. (2012), Nguyen (2012), Cadot et al. (2013), Aeberhardt et al. (2014), Timoshenko (2015), Araujo et al. (2016), Cebreros (2016), Carrère and Strauss-Kahn (2017), Ruhl and Willis (2017), Arkolakis et al. (2018), Li (2018), Berman et al. (2019), Lawless and Studnicka (2019), and Esteve-Pérez (2021), Fitzgerald et al. (2019), Piveteau (2021) or Alborno et al. (2023).

process, where both the firm’s knowledge about foreign markets and its foreign operations grow gradually over time. Economic theories of multinational activity based on the proximity-concentration trade-off have been able to incorporate uncertainty (e.g. Ramondo et al. (2013); Fillat and Garetto (2015)) and learning in the spirit of the Uppsala model (Conconi et al., 2016a). Gumpert et al. (2020) study MNC and exporter dynamics in a proximity-concentration model that incorporates time-varying firm productivity and sunk affiliate setup costs. They use the model to match the observed life-cycle dynamics of exports and affiliates’ sales by new exporters and (new) MNCs, respectively, in the case of France, Germany and Norway. In particular, they rationalise the observed decline in new exporter MNC exit probabilities by exiting from exporting in order to set up an affiliate in the export market recently entered. Finally, examining rare data on sales forecasts of Japanese MNC affiliates, Chen et al. (2023, 2022) find a marked age dependence on affiliate sales forecast errors, e.g. Chen et al. (2023) find that learning can account for 20-40% of the decline in these forecast errors. Although we do not have data on affiliate sales, we find that export sales of French MNCs display spell age dependence. Hence, even if MNC firms are large and experienced, they continue to face sales uncertainty, quickly boosting sales and expanding their product scopes across countries even after having previously and successfully sold in other export destinations.

2 Data and Descriptive Statistics

2.1 Data

Throughout, we exploit merged data from four different sources: the universe of export transactions from French customs data, French firms balance sheet data (FICUS), French firms financial linkages data (LIFI), and CEPII Gravity database. The sample period spans 1993 to 2006. The year 1993 is used to identify existing exporters, allowing us to focus on new exporters by excluding all firms that exported in 1993.⁴

We merge the French customs data with Gravity database and retain only the observations that can be matched. These observations are then matched with the FICUS balance sheet

⁴In Table 1, we include existing exporters to illustrate the difference between age-1 new exporters and all other exporters.

data, keeping only the matched observations. We further exclude firms in the distribution sectors, specifically the two-digit NACE sectors 50, 51, and 52. Additionally, the LIFI dataset is used to identify multinational firms (MNCs), defined as those either owned by a foreign firm or with foreign affiliates. We exclude firms that have ever been owned by a company located in the tax-haven list of FSF-IMF 2000. It is likely that these firms are actually French and not multinational, and that their primary motivation is to avoid French taxes.⁵ We obtain a sample of 2 440 736 firm-country-product-year (*ijpt*) observations for growth and exit. The sample size of conditional entry is 68 936 933 as we expand the dataset to include the never-entry firm-country-product triplets.⁶

In Appendix A.2, we provide the definitions of our dependent variables, namely, growth, exit, and entry. In the exit sample, the year 2006 is used to identify whether export spells are right-censored or failed by the end of 2005. The mean (median) value of growth in the growth sample is 0.0595 (0.0443); the mean (median) value of exit in the exit sample is 0.4183 (0); and the mean (median) value of entry in the entry sample is 0.0033 (0).

2.2 Firm Characteristics of New Exporters

In this subsection we report several firm-level characteristics of different types of exporters. Because new exporters have been shown to drive aggregate exports over the medium to long term, we focus on them. However, as Blum et al. (2020) show that there is an important subgroup of new exporters that behaves very differently to the rest, we pay particular attention to different sub-groups of new exporters susceptible to behave differently.

New exporters are firms that export for the first time during our sample period from 1993 to 2006 and the year 1993 is used to identify old exporters.⁷ If a firm exits the export market

⁵The FSF-IMF 2000 list is a classification of offshore financial centres created by the Financial Stability Forum and IMF to identify jurisdictions with significant offshore financial activity and potential cross-border financial stability risks. It includes Andorra, Anguilla, Antigua and Barbuda, Aruba, Bahamas, Bahrain, Barbados, Belize, Bermuda, British Virgin Islands, Cayman Islands, Cook Islands, Costa Rica, Cyprus, Dominica, Gibraltar, Grenada, Guernsey Sark & Alderney, Hong Kong, Ireland, Isle of Man, Jersey, Lebanon, Liechtenstein, Luxembourg, Macao, Malaysia (Labuan), Malta, Marshall Islands, Mauritius, Monaco, Montserrat, Nauru, Netherlands Antilles, Niue, Palau, Panama, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines, Samoa, Seychelles, Singapore, Switzerland, Turks & Caicos Islands, and Vanuatu.

⁶We discuss the construction of the conditional entry sample in detail in section 3.2.2.3.

⁷In Blum et al. (2020), new exporters are defined in a firm-year-specific manner, where a new exporter in year t is a firm that did not export in year $t - 1$. To construct a table comparable to that of Blum et al. (2020), Table 1 reports the characteristics of age-1 new exporters, which is also a firm-year-specific definition, i.e., the first-ever year of exporting for new exporters.

and then reenters, the reentry year is not treated as the first year of exporting. Ever-MNC and non-ever-MNC firms can be identified from the financial linkage dataset LIFI between 1993 and 2006. A firm is classified as ever-MNC if, at any point between 1993 and 2006, it was owned by a foreign firm or had affiliates in foreign countries. For conciseness, we will interchangeably use the terms MNC and non-MNC in lieu of ever-MNC and non-ever-MNC respectively. To define a new firm and an existing firm, we use the same definition as in Blum et al. (2020). The FICUS dataset contains information on firms' creation dates.⁸ Existing firms are those created two or more years before entering the export market. New firms are then those for which the year of creation is at most one year prior to the year they begin exporting (or Born-Global firms, BGs). Table 1 reports a similar set of firm characteristics as in Table 1 of Blum et al. (2020) by taking the average across years during the 1994-2006 period. The values for exports, sales, and capital are reported in thousands of euros.

Like in Blum et al. (2020), age-1 new exporters account for a tiny share of all exports (around 1.60%) and are smaller in all dimensions, when compared to experienced exporters. Among all exporters, MNC firms represent a relatively small share of the total number of firms (around 13.7%), but they account for almost three-fifths in terms of overall export value, being larger across all dimensions than their non-MNC counterparts. Among age-1 new exporters, we distinguish between two different sets of mutually exclusive categories: MNCs versus non-MNCs, and new firms (BGs) versus existing firms. The number of age-1 new exporters among MNCs is more than ten times smaller than among non-MNCs. However, they are much larger in terms of export value, sales, labor force, and capital, as well as more productive (higher value added per worker). Comparing MNCs with MNC age-1 new exporters, MNC exports per firm are significantly higher, indicating that they do gradually grow. The descriptive evidence is thus indicative of substantial uncertainty affecting also MNCs.

Turning now to BGs and non-BGs age-1 new exporters, BGs represent a small share of all age-1 new exporters, with exports per firm and export intensity being significantly higher than for non-BGs age-1 new exporters. If we include BGs of all ages, rather than restricting the sample to age-1 firms, their share of total exports is 8.476%. This justifies the quantitative importance of studying the dynamics of BGs. Relative to the Chilean new firms reported in

⁸For a tiny share of firms, we don't have information on their date of creation, and we cannot define whether they are new or existing firms.

Blum et al. (2020), French born-global firms (BGs) show lower performance in terms of labor, capital, and value added per worker. French age-1 new exporter BGs are also smaller, less capital intensive and less productive than age-1 new exporter MNCs, with a share of exporters of 2.8% compared to 1% respectively. Age-1 new exporter BGs represent a larger share of new exporters than MNC new exporters do.⁹

Table 1: Firm-level characteristics by exporter type

	All Exporters					Age 1 New Exporters			
	All	Non Age-1 New	Age-1 New	Ever-MNC	Non-MNC	Ever-MNC	Non-MNC	Born Global	Non Born Global
Number of firms	45,651	37,677	7,973	6,266	39,385	453	7,520	1,284	6,471
Share of exporters	1.000	0.8238	0.1762	0.1374	0.8626	0.0100	0.1662	0.0282	0.1428
Total exports	1.82×10^8	1.79×10^8	2.81×10^6	1.05×10^8	7.65×10^7	1.25×10^6	1.56×10^6	1.28×10^6	1.49×10^6
Share of all exports	1.000	0.9840	0.0160	0.5793	0.4207	0.0072	0.0088	0.0075	0.0083
Exports per firm (median)	65	115	6	1,133	46	22	6	13	5
Exports per firm (mean)	3,971	4,721	363	16,761	1,938	2,815	215	989	239
Total sales per firm (median)	1,961	2,473	573	12,419	1,534	4,658	521	394	627
Total sales per firm (mean)	20,036	23,233	4,656	76,602	11,038	26,691	3,354	5,939	4,458
Domestic sales per firm (median)	1,630	2,021	536	8,706	1,299	4,244	490	326	598
Domestic sales per firm (mean)	16,066	18,512	4,293	59,841	9,100	23,876	3,139	4,950	4,219
Export intensity (median)	0.0498	0.0646	0.0143	0.1243	0.0440	0.0071	0.0146	0.0466	0.0113
Export intensity (mean)	0.1566	0.1712	0.0893	0.2367	0.1439	0.0956	0.0889	0.1639	0.0738
Labor (median)	15	19	5	68	12	30	5	3	6
Capital (median)	184	236	57	2,057	138	1,058	51	32	64
Value added per worker (median)	43	44	39	52	42	48	39	33	40

Notes: The sample covers all firms exported in 1994–2006 period and includes firms that exported in 1993. The left part of the Table reports annual average firm characteristics for all exporters in the sample. The right part of the Table reports firm characteristics of New Exporters in their very first year of exporting (age-1, thus excluding re-entrants). An ever-MNC is a firm that, at any point between 1993 and 2006, was owned by a foreign firm or had foreign affiliates. Among new exporters, a Born Global firm is one created at most one year before starting exports, while a Non Born Global Firm is one created two or more years before exporting. Exports, sales, and capital are expressed in thousands of euros. Export intensity is defined as exports divided by total sales.

How different are MNCs from BGs? Table 2 shows that they are not 'the same firm', reporting the corresponding numbers for the four possible exclusive categories of new exporters. Notably, less than 25 percent of all MNCs (5,832) are new firms (1,380 BGs), with only around 8 percent of all new firms (16,692) being MNCs (1,380). Even if they are different, do they display similar new exporter dynamics? All along we will be separately keeping track of MNCs and non-MNCs, as well as new (BGs) and existing firms. Next, we study the new exporter dynamics of MNCs, and after we look at the corresponding dynamics of new firms (BGs).

⁹In appendix B, we present the same table using Blum's definition of new exporters, where a new exporter in year t is the one that did not export in $t - 1$. Therefore, re-entrants are treated as part of the new exporters. The two tables are similar.

Table 2: New exporters: MNCs $\times$ BGs

	Existing Firms	BGs	Total
Non-MNCs	79 676	15 312	94 988
MNCs	4452	1380	5832
Total	84 128	16 692	100 820

Notes: The sample covers 1994–2006. A new exporter is a firm that begins exporting for the first time during the sample period. Among new exporters, an (ever-) MNC is a firm that was at any point owned by a foreign firm or had foreign affiliates between 1993 and 2006. A new born-global firm (BG) is one created at most one year before starting to export, while an Existing Firm is one created two or more years before starting to export.

3 Dynamics of Newly Exporting Multinationals

Several studies have found age-dependence in individual exports, conditional on size (see references in footnote 3). There is also evidence of age dependence in affiliate sales of Multinational Companies (MNCs).¹⁰ These findings contrast with the traditional view of MNCs’ advantage over domestic producers and non-MNC exporters, according to which MNCs’ uncertainty about performance in foreign markets is fully resolved before entry into exporting due to their accumulated experience abroad. It is therefore unclear which mechanisms able to generate age-dependence in exports (such as learning, credit or capacity constraints) could also apply to those firms.

In this Section, we examine whether newly exporting MNCs exhibit the same dynamics as non-MNC new exporters. We start by providing descriptive statistics on the number of destinations, the number of products, exporting values and exit rates by exporting age. We then analyze age dependence in growth and exit dynamics within firm-product-destination export spells, as well as the dynamics across those spells, comparing growth, exit and entry in early versus late spells.

¹⁰For instance, Chen et al. (2023) find that the errors made by Japanese MNCs to forecast affiliate sales decrease significantly with age. Gumpert et al. (2020) show that exit rates of French and Norwegian MNCs (from serving their host market) decrease with age, even though that age-dependence is less pronounced than for exports.

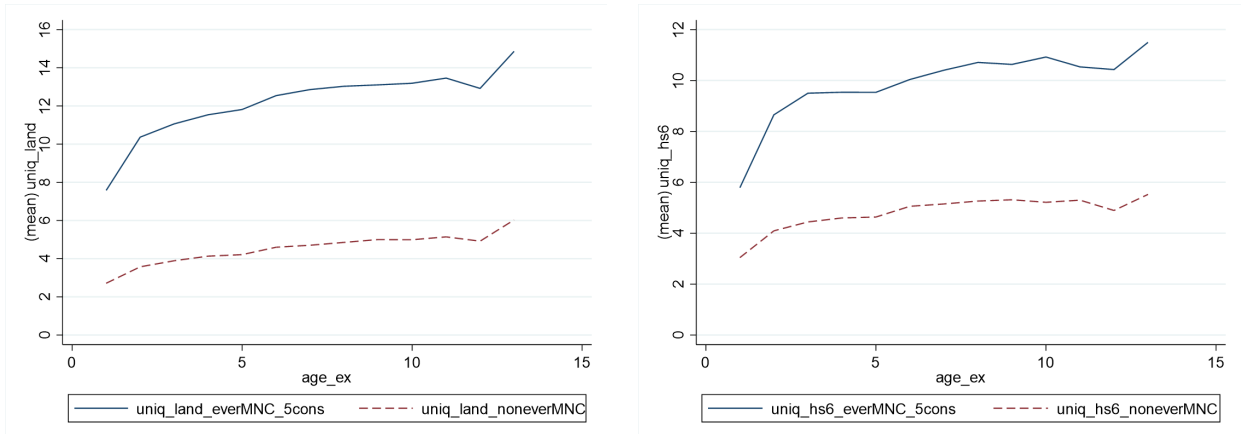
3.1 Descriptive Statistics

We start by describing export dynamics by exporting age.¹¹ We exclude from the analysis all firms that export in our first sample year (1993), and therefore all left-censored spells. In this subsection, a firm is classified as MNC (value = 1) if it ever had a foreign affiliate or was owned by a foreign entity between 1993 and 2006 according to LIFI data. We restrict the analysis to exporters with spell length of *five consecutive years or more*. The goal is to disentangle pure age effects from composition effects due to differences in spell lengths across different types of firms.¹²

3.1.1 Number of countries and products

Figure 1 plots the mean number of destinations (left panel) and HS6 products (right panel) by MNCs (solid line) and non-MNCs (dashed line) exporting ages, respectively. MNCs serve approximately twice as many countries and products over their exporting lifetimes than non-MNCs do. Despite these differences, age patterns appear very similar qualitatively: numbers grow rapidly at first, only to slow down and start plateauing between exporting ages 5 and 10.¹³

Figure 1: Number of export destinations (left) and export products (right), by age and multinational status



Notes: The sample covers the period from 1994 to 2006 for new exporters active in 5 consecutive years (or more). Destinations are defined at the country level; products are defined at the HS6 level. MNC new exporters are represented by a solid blue line, while non-MNC new exporters are represented by a dashed red line.

¹¹Data sources and key variables are presented in Appendix A.

¹²For the sake of completeness, Appendix C.1 shows results on the whole sample.

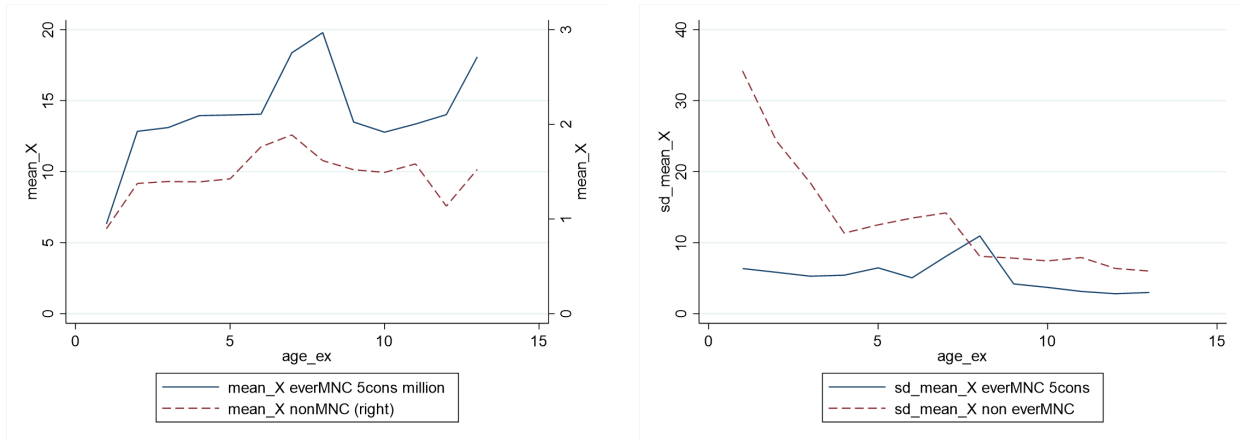
¹³Note that dynamics between ages 1 and 2 might be affected by partial-year effects, as some firms may start exporting late in the calendar year. That will also be the case when we look at within spell growth dynamics, but not when we will examine within spell exit or age-dependence across spells.

3.1.2 Export values: level and dispersion

In addition to adding products and countries gradually, do MNCs also gradually increase export values? Figure 2 documents the evolution of the level (left panel) and dispersion (right panel) of individual exports for MNCs (solid line) and non-MNCs (dashed line) by firm exporting age. The left panel shows average firm exports in million euros, by age and multinational status. Due to an almost 10-fold difference in scale, we use two different y-axes: the left axis corresponds to MNCs, while the right axis corresponds to other firms. In both groups, the age pattern mimics that of the numbers of products and countries: a sharp increase in export value at early ages, after which the export value stabilizes.

The right panel of Figure 2 plots the ratio of the standard deviation in firm exports divided by its mean (coefficient of variation), by exporting age and multinational status. We find a broadly downward-sloping relationship between that measure of dispersion and age, both among multinationals and non-multinationals.¹⁴ Because the reduction appears stronger among the latter, a form of convergence in individual exports over time is apparent.

Figure 2: Export value by age and multinational status: mean (left panel) and normalized standard deviation (right panel).



Notes: The sample covers the period from 1994 to 2006 for new exporters active for 5 consecutive years (or more). Mean export values (left panel) are expressed in millions of euros at the firm level. The normalized standard deviation (right panel) is defined as the average across firms in each group of the coefficient of variation (standard deviation to the mean ratio) by exporting age. The standard deviation is computed across firms by exporting age and multinational status. MNC new exporters are represented a solid blue line, while a dashed red line captures non-MNC new exporters.

Figure 2 shows that MNCs sell more and face less uncertainty than non-MNCs. Export values

¹⁴The 'rebound' after five years likely comes from our sample construction. As noted above, we only consider firms exporting for five consecutive years.

increase and uncertainty decreases in both groups of new exporters as they become more experienced. But these apparent similarities in exporting age patterns might be masking different growth trends across groups. To gain further insight, we run a simple regression of *log* firm export values on exporting age, allowing for firm-specific differences in linear trends:

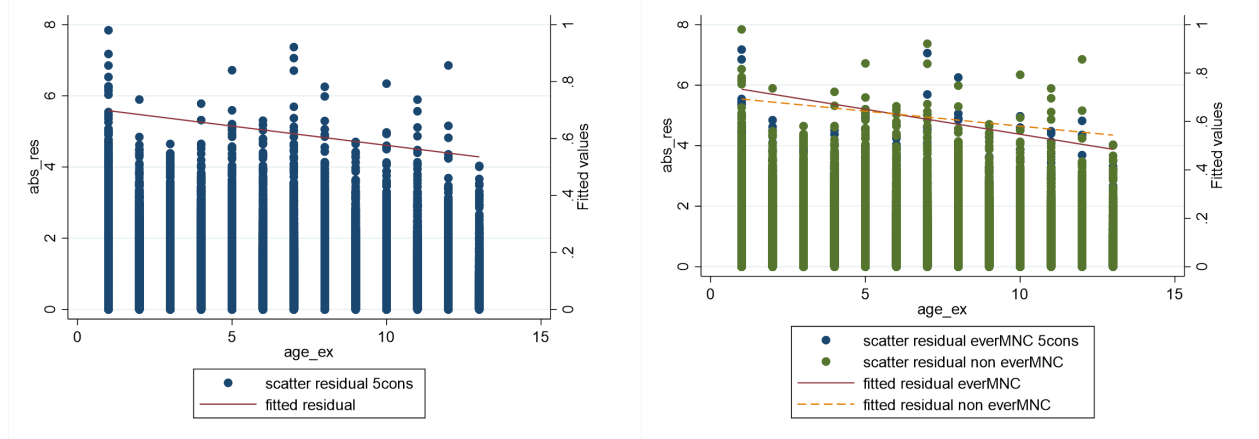
$$\ln(\text{export value})_{it} = \alpha_i + \beta_i * \text{exporting age}_{it} + \epsilon_{it}, \quad (1)$$

The residual ϵ_{it} thus captures percentage deviations in firm i 's exports from their individual growth trend at different exporting ages, t . We examine the hypothesis that these residuals (in absolute value) fall with age, which would be consistent with a firm-level learning process.

Figure 3 plots the absolute value of the estimated *residuals*, $|\hat{\epsilon}_{it}|$ from (1) and a line of best fit (on a different right-hand scale). The left panel reports results for all firms while the right panel does so by MNC status (MNCs vs non-MNCs). In both groups of firms, we observe a negative relationship between the absolute value of the estimated residual and exporting age. This indicates that more experienced exporters converge toward their individual growth trends (left and right panels), and that that convergence is faster among MNCs (right panel).

We thus find that both non-MNCs and MNCs display similar qualitative growth patterns. They both enter below scale with few products and into few markets. Conditional on surviving, they rapidly expand at all margins as they progressively unveil the surrounding uncertainty. Therefore, it appears that also MNCs face uncertainty about their performance in foreign markets and/or significant frictions to reaching their optimal foreign sales.

Figure 3: Residuals (in absolute value) from a regression of $\log$ export value on age, and fitted trend by exporting age: all firms (left panel) and by MNC status (right panel)



Notes: The sample covers the period from 1994 to 2006 for new exporters active for 5 consecutive years or more. The residuals are plotted in absolute value from (1) (left axis), based on a regression of the logarithm of export value on exporting age at the firm level, with firm-specific slope coefficients and firm fixed effects. The right axis shows the fitted trend values. The left panel is for all firms, while the right panel separates MNCs (solid red line) and non-MNCs (dashed orange line).

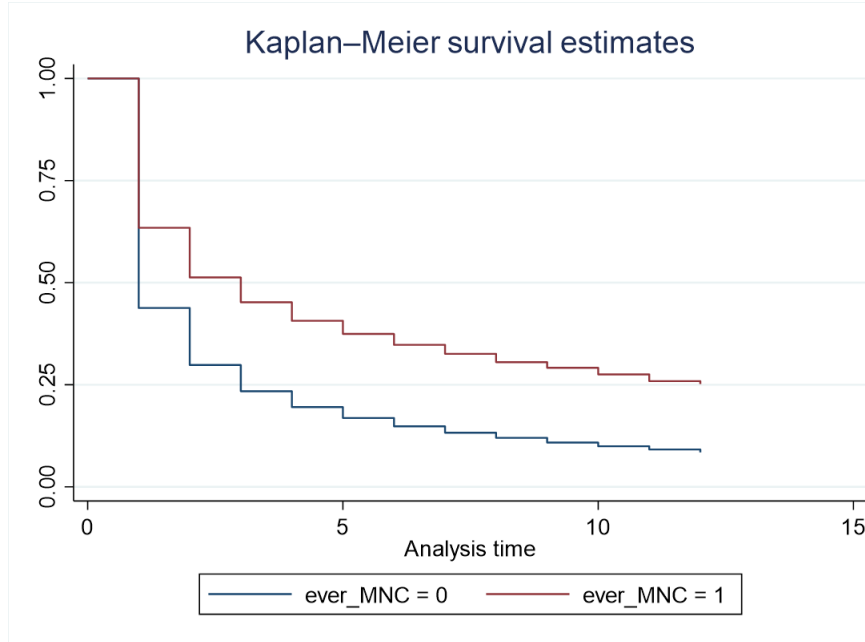
3.1.3 Survival/hazard rates

We now consider exit (from exporting) by age and multinational status. In what follows we define a firm's exit in year t as the absence of exports in year $t + 1$ despite positive exports in year t .¹⁵

Figure 4 plots the survival rates for both MNCs and non-MNCs. The survival rate declines sharply at first, only to stabilize as firms gain more experience in the export market. The survival rate of MNCs remains higher than that of non-MNCs at any given age.

¹⁵This definition implies that exiters may later re-enter export markets. In a robustness check we use a more stringent definition of exit, where only permanent exit counts as failure. In other words, the exit hazard occurs in the last-ever export year of the firm. Figures are omitted to save space, but available upon request. We still find negative age-dependence among both ever-MNCs and non-ever MNCs. In particular, more than half of MNC export starters have permanently exited after four years.

Figure 4: Kaplan-Meier cumulative export survival rates by exporting age and multinational status



Notes: The sample covers the period from 1994 to 2006. Exit is a binary variable that takes value 1 if exports are positive in year t and zero in year $t + 1$. We use the *sts* command in Stata to produce the graph. The survival curve for MNCs is solid red, and the survival curve for non-MNCs is solid blue.

Altogether, the uncovered descriptive evidence reveals a consistent picture. New exporter MNCs, while more successful at exporting, have qualitatively similar exporting age patterns as non-MNC new exporters, indicating that either uncertainty and/or frictions are also prevalent among MNCs. Exit rates are high at the beginning of their export tenure, but conditional on surviving, they quickly expand the number of products, destinations, and overall export values. After approximately 5 years of exporting, all of those typically stabilize (around higher levels and trends than for non-MNC firms).

3.2 Regression Results

We now report regression results where we seek to isolate age effects from other determinants of export dynamics. Our empirical investigation of export dynamics features three main dependent variables: export growth, exit, and conditional entry. All three variables are measured at the firm-country-product-year (*ijpt*) level.¹⁶

¹⁶See Appendix A for additional details on the definition of these variables.

3.2.1 Within-spell dynamics

We first consider within-spell export dynamics, where a spell denotes a firm-country-product triplet, ijp . We want to establish whether growth and exit rates vary systematically with spell age, controlling for other determinants. Spell age is defined as the current exporting year for a given spell minus the first year in which the spell started exporting, plus one. We group spell ages equal or above 8 into an 'age 8' category.

3.2.1.1 Growth regressions. We estimate the following linear export growth model:

$$\Delta \log X_{ijpt} = \alpha_0 + \mathbf{age}_{ijpt}' \alpha + (\mathbf{age}_{ijpt} \times MNC_{it})' \alpha_{\mathbf{mnc}} + X_{it}' \beta + G_{jt}' \gamma + \eta_t + \eta_{ijp} + u_{ijpt}, \quad (2)$$

where α_0 is a constant and $\mathbf{age}_{ijpt}$ is a vector of dummies for the following exporting age bins: 2, 3, 4, 5, 6, and 7. There is no foreign sales growth at age 1, and α_0 captures the growth rate at age 8 or older, capturing the average growth rate of experienced exporters. Therefore, the vector α captures the difference between the growth rate at current age and the growth rate of old exporters, i.e. 'age 8 or older'. MNC_{it} equals one if, in the current year, the firm is a multinational. X_{it} is a set of firm-level variables, and G_{jt} is a set of gravity variables at the destination-year level.¹⁷ η_{ijp} denotes a firm-country-product (or spell) fixed effect and η_t denotes the year fixed effect. $\alpha_{\mathbf{mnc}}$ captures the differential variation in export growth over the duration of an MNC export spell relative to a non-MNC export spell.

Previous work has shown that young (exporting) firms exhibit higher growth rates, conditional on size (Alessandria et al., 2021b; Albornoz et al., 2012, 2023; Berthou and Vicard, 2015). Here we test the hypothesis that multinational companies face similar age-dependent dynamics, namely that $\alpha_{\mathbf{mnc}} = \mathbf{0}$.

Table 3 presents within-spell growth results using two specifications. Compared to specifica-

¹⁷The firm-level variables are total factor productivity (TFP) growth, group affiliation, lagged log total sales (in million euros), and within-year leverage ratio tercile. TFP is calculated at the firm level using the method proposed by (Levinsohn and Petrin, 2003). See Appendix A.2.5 for further details on TFP estimation. The gravity variables are population weighted distance to France, population, GDP per capita, contiguity with France, common official language, colonial ties (post 1945 and ever), GATT/WTO membership, Regional Trade Agreements with the EU, common legal origin, common currency and participation in cooperation agreements between the EU and African, Caribbean and Pacific countries. We use the same firm-level variables and gravity variables in all the regressions sections in this paper. See Appendix A.2.5 for variable definitions and their simple descriptive statistics.

tion (2) (column 2), the results under column 1 come from a specification that excludes the interaction between age dummies and the MNC dummy, only included on its own to capture differences in growth levels between MNCs and non-MNCs. In both specifications, the growth rate is high at the beginning of the spell, and decreases in spell age. By spell age 7, the growth rate appears to have converged to its steady-state, i.e. the growth rate at age 8 or more of old exporters. Interestingly, no significant difference is found between the growth patterns of MNCs and non-MNCs, as all α_{mnc} coefficients appear statistically non-significant. Therefore, MNC new exporting firms exhibit an export growth pattern similar to their non-MNC counterparts, with faster growth early in their exporting spells, reaching their steady states after 5-7 years.

Table 3: MNC Growth Regressions Within Spell

	(1) Growth		(2) Growth	
	Coef.	SE	Coef.	SE
age2	0.639	(0.038)***	0.636	(0.040)***
age3	0.151	(0.032)***	0.153	(0.034)***
age4	0.080	(0.027)***	0.090	(0.029)***
age5	0.052	(0.023)**	0.061	(0.026)**
age6	0.036	(0.018)**	0.040	(0.020)**
age7	0.012	(0.014)	0.015	(0.016)
MNC_age2			0.014	(0.038)
MNC_age3			-0.004	(0.030)
MNC_age4			-0.034	(0.029)
MNC_age5			-0.029	(0.034)
MNC_age6			-0.011	(0.032)
MNC_age7			-0.004	(0.031)
MNC_age8			0.005	(0.030)
MNC_it	-0.009	(0.026)		
_cons	0.219	(0.102)**	0.215	(0.103)**
gravity		yes		yes
firm level		yes		yes
year FE		yes		yes
spell FE		yes		yes
<i>N</i>	687,616		687,616	
<i>R</i> ²	0.201		0.201	

Notes: The sample covers the period from 1994 to 2006. The OLS dependent variable is export growth, defined in appendix A.2.2, and computed from age 2. The constant term captures growth at exporting age 8 or older, while the age dummies capture deviations from growth at that age. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables.

3.2.1.2 Exit regressions. We next examine whether new MNC exporters face a decreasing probability of exit by spell age, similar to the probability of their non-MNC counterparts. To

do so, we estimate the following discrete-time duration model:

$$hazard_{ijpt} = 1 - \exp \left[-\exp \left(\mathbf{age}_{ijpt}' \alpha + (\mathbf{age}_{ijpt} \times MNC_{it})' \alpha_{\text{mnc}} + X'_{it} \beta + G'_{jt} \gamma + \eta_t + \eta_{ijp} + u_{ijpt} \right) \right], \quad (3)$$

Like in the above growth regressions, $\mathbf{age}_{ijpt}$ is a vector of dummies for the following exporting age bins: 1, 2, 3, 4, 5, 6, 7, and 8. Age 8 captures the hazard at exporting age 8 or older. Therefore, the vector α captures the baseline hazard value across different spell ages. η_t denotes the year fixed effect. Unlike in growth regressions, η_{ijp} denotes a firm-country-product (spell) random effect. We use two different random effect distributions. For results under columns 1 and 2, η_{ijp} follows a normal distribution, while under columns 3 and 4, η_{ijp} follows a gamma distribution (as a robustness check).

Table 4 reports estimation results. Columns 2 and 4 report estimates of the model specified in (3), while under columns 1 and 3, estimates of a similar specification but without interactions between spell age and MNC status are reported. In the latter, only the MNC dummy is present to detect differences in spell age average exit rates by MNC status. In all columns coefficients are reported in exponential form: a coefficient above (below) one indicates a higher (lower) exit probability than non-MNCs at the same spell age.

First, we find decreasing age coefficients across all specifications. The hazard drops sharply at the beginning of the spell, but becomes relatively stable from age 5 onwards. This pattern is consistent with the Kaplan-Meier unconditional cumulative export survival rates by exporting age and multinational status displayed in Figure 4. Regarding the coefficients of interaction between MNCs and age dummies, MNCs exhibit a lower hazard rate at any given age compared to other new exporters. But just like non-MNCs, they are less likely to exit as they become more experienced. These findings appear to be robust to allowing for right-tail unobserved firm heterogeneity, since they apply equally for the two different distributions of random effects.

Table 4: MNC Exit Regressions Within Spell

	(1)	(2)	(3)	(4)
	Exit	Exit	Exit	Exit
age1	15.373 (0.715)***	15.248 (0.709)***	16.892 (0.794)***	16.836 (0.791)***
age2	6.915 (0.327)***	7.012 (0.331)***	7.582 (0.362)***	7.738 (0.370)***
age3	5.706 (0.271)***	5.819 (0.276)***	6.263 (0.301)***	6.430 (0.309)***
age4	5.160 (0.246)***	5.297 (0.252)***	5.667 (0.273)***	5.860 (0.283)***
age5	4.717 (0.226)***	4.836 (0.232)***	5.184 (0.251)***	5.355 (0.260)***
age6	4.596 (0.221)***	4.733 (0.228)***	5.054 (0.246)***	5.244 (0.256)***
age7	4.306 (0.208)***	4.437 (0.215)***	4.736 (0.232)***	4.918 (0.242)***
age8	4.104 (0.197)***	4.121 (0.198)***	4.513 (0.219)***	4.568 (0.223)***
MNC_age1		0.958 (0.005)***		0.959 (0.005)***
MNC_age2		0.852 (0.009)***		0.851 (0.009)***
MNC_age3		0.828 (0.011)***		0.826 (0.011)***
MNC_age4		0.800 (0.012)***		0.797 (0.012)***
MNC_age5		0.804 (0.014)***		0.802 (0.014)***
MNC_age6		0.787 (0.017)***		0.784 (0.016)***
MNC_age7		0.784 (0.020)***		0.781 (0.020)***
MNC_age8		0.895 (0.017)***		0.890 (0.017)***
MNC_it	0.902 (0.004)***		0.903 (0.004)***	
gravity	yes	yes	yes	yes
firm level	yes	yes	yes	yes
year FE	yes	yes	yes	yes
spell RE	normal	normal	gamma	gamma
N	1502022	1502022	1502022	1502022

Notes: The sample covers the period 1994–2006. Year 2006 identifies left-censored spells. The dependent variable is the exit hazard, defined in appendix A.2.3. Reported coefficients obtain from estimation of discrete-time proportional hazard models with random effects (RE). Normal and gamma-distributed random effects follow from `xtcloglog` and `pgmhaz8`, respectively. “Age 8” refers to exporting age 8 or older. Appendix A.2.5 contains the list and definitions of gravity and firm-level variables. Standard errors are in parentheses. * p<0.10, ** p<.05, *** p<.01

3.2.2 Dynamic complementarities across markets

Having identified similar age effects in growth and exit probabilities of MNCs' and non-MNCs' product-markets, consistent with uncertainty and/or frictions being prevalent within spells, we now turn to look into age effects that spillover across product-markets for both sets of firms. To do so, we examine systematic differences between age dynamics in a firm's first country-product spell relative to later spells involving other countries or products. The idea is to detect potential dynamic complementarities across country-product markets. These may come from, for instance, a customer base accumulation process that is common across countries and/or products, or more likely, from uncertainty about the ability of the firm to export profitably across product-markets (see e.g. Albornoz et al. (2023)). We build on the latter, adding interaction terms between age effects and multinational status. Our goal is to test whether multinational firms exhibit the same form of age dependence in exports as non-multinationals, controlling for differences in size, productivity growth, as well as other relevant drivers of export dynamics.

3.2.2.1 Growth regressions We estimate the following linear model:

$$\begin{aligned}\Delta \log X_{ijpt} = & \alpha_0 + \mathbf{Age}'\alpha + (\mathbf{Age} \times MNC_{it})' \alpha_{\mathbf{mnc}} + \alpha_{YMP} (FY_{ijp,t-1} \times FM_{ij} \times FP_{ip}) \\ & + \alpha_{YMPmnc} (FY_{ijp,t-1} \times FM_{ij} \times FP_{ip}) \times MNC_{it} \\ & + \mu MNC_{it} + X'_{it}b + G'_{jt}c + \{FE\} + u_{ijpt},\end{aligned}\tag{4}$$

where $\mathbf{Age} = (FY_{ijp,t-1}, FM_{ij}, FP_{ip}, FY_{ijp,t-1} \times FM_{ij}, FY_{ijp,t-1} \times FP_{ip}, FM_{ij} \times FP_{ip})$.¹⁸ In Equation (4), $\Delta \log X_{ijpt}$ is firm i 's growth rate of the value of product p exports to market j between t and $t - 1$, $FY_{ijp,t-1}$ is a dummy indicating whether firm i exported product p to destination j in $t - 1$ for the first time, FM_{ij} indicates whether j is firm i 's first export market and FP_{ip} takes value 1 if p is firm i 's first ever exported product. MNC_{it} is a binary variable equal to one if firm i is either foreign-owned or owns foreign affiliates. G_{jt} is a vector of gravity variables. X_{it} includes controls for other potential firm-specific drivers of export dynamics: TFP growth, leverage (dummy variables for the top and middle terciles), lagged log total sales

¹⁸Exporters do not necessarily export the same set of products in each initial market during their first year of exporting. To capture the first country-product spell effect, the dummy variable $FM_{ij} \times FP_{ip}$ is constructed based on the first country-product pair. This definition is used throughout the rest of the paper.

in million euros ($L1lncatotal$) and group ownership. $\{FE\}$ denotes a large set of fixed effects, described below.

$\alpha_{YMP} + \alpha_{YMPmnc} \times MNC_{it}$ captures the differential age patterns in the first export spell sales growth of new exporting MNCs, relative to their growth later, in subsequent products and/or markets. Below, we test whether this sum of coefficients is nonnegative. $\alpha_{YMP} > 0$ indicates higher non-MNC export growth at the beginning of the first export spell relative to the beginning of other export spells. When $\alpha_{YMP} = 0$, there is no additional age dependence in the first spell relative to later export spells: experience from past spells does not affect growth dynamics.

Table 5 presents the results of specification (4) growth regressions. In all columns we control for firm and country (gravity) characteristics, as well as year dummies and country dummies. Odd-numbered columns include product fixed effects, while even-numbered ones include firm fixed effects. Under columns 3 and 4, the coefficient α_{YMP} is significantly positive across all specifications: for instance column 3 reports a premium of almost 16 percentage points in growth rates in the first year in the first market with the first product. However, the interaction term α_{YMPmnc} is not significant, implying the same differential age patterns in growth among MNCs and non-MNCs. Under columns 5 and 6, we replace MNC_{it} with two distinct variables capturing whether the French exporter is foreign-owned or is a parent with foreign affiliates. Results are similar except for $\alpha_{YMP} \times Parent$ when using product fixed effects. The sum of the coefficients of interest remains significantly positive. Overall, growth rates of multinational new exporters show the same age-dependence as that of non-multinational ones, conditional on size and firm characteristics. If anything, the identified age-dependence is even more pronounced for French parent firms.

Table 5: MNC Growth Regressions Baseline

	(1)	(2)	(3)	(4)	(5)	(6)
	Growth	Growth	Growth	Growth	Growth	Growth
α_{YMP}	0.170 (0.023)***	0.223 (0.024)***	0.158 (0.025)***	0.228 (0.028)***	0.158 (0.025)***	0.228 (0.028)***
MNC_it	0.000 (0.012)	-0.009 (0.022)	-0.019 (0.017)	-0.029 (0.029)		
α_{YMPmnc}			0.042 (0.058)	-0.017 (0.053)		
α_{YMPfo}					0.031 (0.060)	-0.027 (0.053)
α_{YMPpa}					0.273 (0.160)*	0.201 (0.151)
FO_it					-0.019 (0.016)	-0.019 (0.031)
Parent_it					-0.024 (0.059)	-0.049 (0.053)
gravity	yes	yes	yes	yes	yes	yes
firm level	yes	yes	yes	yes	yes	yes
year FE	yes	yes	yes	yes	yes	yes
country FE	yes	yes	yes	yes	yes	yes
product FE	yes		yes		yes	
firm FE		yes		yes		yes
Observations	837,634	837,634	837,634	837,634	837,634	837,634
R^2	0.020	0.058	0.020	0.058	0.020	0.058
$\alpha_{YMP} + \alpha_{YMPmnc}$			0.201***	0.211***		
$\alpha_{YMP} + \alpha_{YMPfo}$					0.189***	0.201***
$\alpha_{YMP} + \alpha_{YMPpa}$					0.431***	0.429***

Notes: The sample covers the period from 1994 to 2006. The OLS dependent variable is export growth, defined in Appendix A.2.2. Growth rates are computed from age 2. In specifications (1), (3), and (5), we use product fixed effects (FEs), and in specifications (2), (4), and (6), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables. * p<0.10, ** p<.05, *** p<.01

3.2.2.2 Exit regressions

We estimate the following linear probability model:

$$\begin{aligned}
Exit_{ijpt} = & \beta_0 + \mathbf{Age}'\beta + (\mathbf{Age} \times MNC_{it})' \beta_{mnc} + \beta_{YMP} (FY_{ijp,t} \times FM_{ij} \times FP_{ip}) \\
& + \beta_{YMPmnc} (FY_{ijp,t} \times FM_{ij} \times FP_{ip}) \times MNC_{it} + \mu MNC_{it} + X'_{it}\beta + G'_{jt}\gamma \\
& + \{FE\} + w_{ijpt},
\end{aligned} \tag{5}$$

where $\mathbf{Age}$ is defined as in specification (4), except that we use $FY_{ijp,t}$ instead of $FY_{ijp,t-1}$. $Exit_{ijpt}$ is a binary variable that takes value one if firm i stops exporting product p to market j in year $t + 1$ or later after having exported in year t , and zero otherwise. $FY_{ijp,t}$ is a binary variable that takes value one if firm i is in the first year of its jp (market-product) spell. FM_{ij} , FP_{ip} , G_{jt} , X_{it} and the fixed effects are defined as for growth regressions. This means that age effects are estimated conditional on size and productivity growth.

A positive β_{YMP} coefficient indicates that new exporter non-MNC exit (ending a spell) is more likely at the beginning of the first export spell than at the beginning of subsequent spells. A positive sum of coefficients $\beta_{YMPmnc} \times MNC_{it} + \beta_{YMP}$ indicates that it is also the case for multinational counterparts.

Table 6 shows the results using exit as the dependent variable. Consider first regressions with product fixed effects under columns 1, 3, and 5. In all columns the coefficient β_{YMP} is significantly positive: exit rates are nearly 16 percentage points higher in the first year of a spell than in later years of a spell or than in the first year of subsequent spells. However, that early exit pattern doesn't apply to multinationals. The sum of coefficients $\beta_{YMPmnc} \times MNC_{it} + \beta_{YMP}$ is non-significant, whether we consider all multinationals or separate parents and affiliates. Under columns 2, 4 and 6 the results for firm fixed effect regressions are reported. In these specifications, identification relies on within-firm comparison between spells. All observations where new exporters serve only a single product in a single market once are thus excluded from the estimation. In that case, there seems to be no difference in exit in the first spell relative to other spells for multinationals, and less exit among Non-MNCs. We conclude that multinationals do not exhibit age-dependence in exit, unlike non-multinationals¹⁹.

One may argue that our definition of exit includes multinationals switching from exports to affiliate sales to serve the same foreign market, as in Gumpert et al. (2020). And indeed, Conconi et al. (2016b) also show that many multinationals set up affiliates after having successfully exported to that market. But in those cases, we would expect exit rates to increase with the time spent in a product-market (spell age). To make sure that our results are not affected by exporters switching status into setting an affiliate, we run similar regressions under a different definition of exit. Namely, for firms that stop exporting to a given country but have an affiliate in that country for some years, we treat these firms as continuing to export to that country during those years. The results, presented in Appendix C.2.2, are broadly similar to those reported in Table 6 .

¹⁹This result also holds in a subsample containing ever multinationals only, as shown in the Appendix C.2.1.

Table 6: MNC Exit Regressions Baseline

	(1)	(2)	(3)	(4)	(5)	(6)
	Exit	Exit	Exit	Exit	Exit	Exit
β_{YMP}	0.142 (0.009)***	-0.036 (0.006)***	0.161 (0.009)***	-0.038 (0.006)***	0.161 (0.009)***	-0.037 (0.006)***
MNC_it	-0.013 (0.007)*	-0.019 (0.013)	-0.010 (0.012)	-0.026 (0.021)		
β_{YMPmnc}			-0.156 (0.032)***	0.001 (0.023)		
β_{YMPfo}					-0.145 (0.038)***	0.001 (0.025)
β_{YMPpa}					-0.158 (0.042)***	-0.001 (0.064)
FO_it					-0.014 (0.012)	-0.036 (0.022)*
Parent_it					0.017 (0.040)	0.017 (0.027)
gravity	yes	yes	yes	yes	yes	yes
firm level	yes	yes	yes	yes	yes	yes
year FE	yes	yes	yes	yes	yes	yes
country FE	yes	yes	yes	yes	yes	yes
product FE	yes		yes		yes	
firm FE		yes		yes		yes
Observations	1,471,412	1,448,886	1,471,412	1,448,886	1,471,412	1,448,886
R^2	0.228	0.309	0.228	0.310	0.229	0.310
$\beta_{YMP} + \beta_{YMPmnc}$			0.005	-0.037		
$\beta_{YMP} + \beta_{YMPfo}$					0.016	-0.037
$\beta_{YMP} + \beta_{YMPpa}$					0.003	-0.038

Notes: The sample covers the period from 1994 to 2006. The OLS dependent variable is export exit, defined in Appendix A.2.3. Year 2006 identifies exit in 2005. In specifications (1), (3), and (5), we use product fixed effects (FEs), and in specifications (2), (4), and (6), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables. * p<0.10, ** p<.05, *** p<.01

3.2.2.3 Conditional Entry regressions We now consider the dynamics of entry into new product-country pairs. Our focus is on *conditional* entry, that is entry conditional on a first successful export spell.

To do that, we build a special sample. The sample excludes old exporters and firms with a single observation, and focuses on successful new exporters. For each of these firms and their products, we square our dataset to consider all possible countries and years, including those without recorded exports. Specifically, if in year t a firm-product pair ip is observed in the original sample for the first time, we define an entry dummy for every country j in the sample where ip was *not* observed in t . We fill in observations in every year following t , until firm i actually enters j with product p (or until 2006, if it does not). Once/if the entry is recorded,

that ijp triplet is dropped from the sample for the subsequent years.²⁰

We estimate the following linear probability model:

$$\begin{aligned}
Entry_{ijpt} = & \gamma_0 + \mathbf{Age}'\gamma + (\mathbf{Age} \times MNC_{it})'\gamma_{mnc} + \gamma_Y FY_{i,t-1} + \gamma_{YM} (FY_{i,t-1} \times FM_{ij}) \\
& + \gamma_{YP} (FY_{i,t-1} \times FP_{ip}) + \gamma_{Ymnc} FY_{i,t-1} \times MNC_{it} \\
& + \gamma_{YMmnc} (FY_{i,t-1} \times FM_{ij}) \times MNC_{it} + \gamma_{Ypmnc} (FY_{i,t-1} \times FP_{ip}) \times MNC_{it} \\
& + \mu MNC_{it} + X'_{it}\beta + G'_{jt}\gamma + \{FE\} + v_{ijpt},
\end{aligned} \tag{6}$$

where now $\mathbf{Age} = (FM_{ij}, FP_{ip}, FM_{ij} \times FP_{ip}, FY_{i,t-1} \times FM_{ij} \times FP_{ip})$; $Entry_{ijpt}$ is a binary variable that takes value one if firm i enters destination j with product p in year t , and zero otherwise. $FY_{i,t-1}$ equals one if $t - 1$ was the first-ever year in the *firm's* export history. FM_{ij} , FP_{ip} , G_{jt} , X_{it} and the fixed effects are defined as above,²¹ implying that age effects are estimated conditional on size and TFP growth. Standard errors are clustered at the firm level.

Table 7 presents the baseline entry results at the firm-market-product level. Successful new exporters are more likely to add products to the same initial country ($\gamma_{YM} > 0$) or to export the same initial product to other countries ($\gamma_{YP} > 0$) immediately after entry, relative to expanding later or with different country-product pairs. Tests of different sums of coefficients show that the same patterns are found among multinational exporters, with a more significantly positive intensity among foreign-owned firms than among parents. This is true irrespective of whether we use product or firm fixed effects.²² Overall, the results point towards gradual export expansion strategies, even among multinationals, favoring familiar countries or products.

²⁰The construction of the sample is explained in detail in the Appendix. We note that squaring the sample for all potential countries and years dramatically increases its size. For computational reasons, we then take a random draw of 30% of the initial sample; this still leaves us with roughly 69 million observations.

²¹By construction of the sample, firm i can only enter in a jp market once. However, in a few cases, a firm starts exporting to different combinations ('swaps') of its initial countries and products in its second year as exporter. Hence γ_{YMP} can be estimated.

²²In a robustness check, we run the same regressions on a sample of ever-multinationals only, and find qualitatively similar patterns. Results are presented in the Appendix C.2.1.

Table 7: MNC Entry Regressions Baseline

	(1) Entry	(2) Entry	(3) Entry	(4) Entry	(5) Entry	(6) Entry
γ_Y	0.00078 (0.00010)***	-0.00008 (0.00011)	0.00076 (0.00010)***	-0.00000 (0.00010)	0.00076 (0.00010)***	-0.00000 (0.00010)
γ_{YM}	0.02566 (0.00170)***	0.02550 (0.00154)***	0.03008 (0.00229)***	0.02976 (0.00212)***	0.03008 (0.00229)***	0.02976 (0.00212)***
γ_{YP}	0.00122 (0.00022)***	0.00121 (0.00020)***	0.00105 (0.00020)***	0.00090 (0.00020)***	0.00105 (0.00020)***	0.00090 (0.00020)***
γ_{Ymnc}			-0.00007 (0.00035)	-0.00052 (0.00038)		
γ_{YMmnc}			-0.01332 (0.00331)***	-0.01275 (0.00300)***		
γ_{Ypmnc}			0.00154 (0.00092)*	0.00221 (0.00078)***		
γ_{Yfo}					-0.00001 (0.00037)	-0.00052 (0.00041)
γ_{YMfo}					-0.01315 (0.00337)***	-0.01263 (0.00307)***
γ_{YPfo}					0.00140 (0.00096)	0.00211 (0.00080)***
γ_{Ypa}					-0.00063 (0.00093)	-0.00073 (0.00085)
γ_{Ympa}					-0.01403 (0.00697)**	-0.01343 (0.00702)*
γ_{YPpa}					0.00354 (0.00246)	0.00374 (0.00273)
MNC_it	0.00050 (0.00017)***	0.00005 (0.00017)	0.00036 (0.00015)**	-0.00016 (0.00019)		
FO_it					0.00033 (0.00016)**	-0.00018 (0.00021)
Parent_it					0.00055 (0.00025)**	-0.00018 (0.00033)
gravity	yes	yes	yes	yes	yes	yes
firm level	yes	yes	yes	yes	yes	yes
year FE	yes	yes	yes	yes	yes	yes
country FE	yes	yes	yes	yes	yes	yes
product FE	yes		yes		yes	
firm FE		yes		yes		yes
Observations	49,411,867	49,411,867	49,411,867	49,411,867	49,411,867	49,411,867
R^2	0.011	0.014	0.011	0.014	0.011	0.014
$\gamma_Y + \gamma_{Ymnc}$			0.001**	-0.001		
$\gamma_{YM} + \gamma_{YMmnc}$			0.017***	0.017***		
$\gamma_{YP} + \gamma_{Ypmnc}$			0.003***	0.003***		
$\gamma_Y + \gamma_{Yfo}$					0.001**	-0.001
$\gamma_{YM} + \gamma_{YMfo}$					0.017***	0.017***
$\gamma_{YP} + \gamma_{YPfo}$					0.002***	0.003***
$\gamma_Y + \gamma_{Ypa}$					0.0001	-0.0007
$\gamma_{YM} + \gamma_{Ympa}$					0.016**	0.016**
$\gamma_{YP} + \gamma_{YPpa}$					0.005*	0.005*

Notes: The sample covers the period from 1994 to 2006. The OLS dependent variable is export entry, defined in Appendix A.2.4. In specifications (1), (3), and (5), we use product fixed effects (FEs), and in specifications (2), (4), and (6), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables. * p<0.10, ** p<.05, *** p<.01

4 Born-Global New Exporter Dynamics

Blum et al. (2020) show that (born-global) new business entities behave much like experienced exporters: they start large and have high survival rates, explaining about 85% of the medium-term new exporters' Chilean export growth.²³ But because Chile is quite different from France in terms of the sector composition of aggregate exports, size, and geographical location, in this section, we identify and examine if French BGs also display experienced-exporter-like (new exporter) dynamics.²⁴

We start by presenting descriptives about BGs, including some comparisons between Chilean and French BGs. In order to assess whether French BGs behave as experienced exporters, we then present descriptives about the dynamics of BGs' export product and geographical scope by BGs' exporting age: the evolution of the number of products, destinations as well as the level and volatility of export values, and exit rates, compared to existing (as opposed to new) French entities that are new as exporters. We then proceed to perform similar empirical exercises as in the previous Section.

4.1 Descriptive Statistics

4.1.1 Difference between Chilean BGs and French BGs

Blum et al. (2020) show that around 70 percent of BGs (new Chilean entities) are owned by existing businesses and are likely the by-product of either domestic spin-offs or foreign direct investment (24 percent). In the LIFI data, we can identify whether a firm belongs to a group and the firm ID of the group head. Among new exporters, we can therefore determine which firms are owned by another firm (i.e., when the group head has a different firm ID). At the firm-year level, 29.5% of BGs are owned by another firm, compared with 23.6% of non-BGs; thus, the share is higher among BGs. For new exporters whose group head is a French firm, we

²³As in Blum et al. (2020) we define BGs as 'firms born at most one year before becoming an exporter... (p.398)'. We use data on firm creation dates from the FICUS/FARE datasets. Namely, if the year of beginning exporting minus the year of firm creation is no greater than one, then the firm is considered a 'new firm'. Whenever a firm's creation date differs across years in the data we assign the most frequent one to all years. We exclude observations with missing creation date or where the export start date comes before the creation date (which account for a tiny share).

²⁴Although we cannot exploit as detailed ownership information as Blum et al. (2020) do, we can nonetheless examine the export dynamics of newly created firms that are also new exporters (BGs).

can use our French data to examine the characteristics of the parent firm. At the firm-year level, among BGs (non-BGs) owned by a French parent firm, the share of parent firms that export is 9.4% (7.4%), and the average export value is 154,203 (181,086) euros. We call these firms "born-global" (BGs). If their export behavior closely resembles that of experienced exporters, it is interpreted as being consistent with their upstream owners having prior export experience.

In our sample, 74.3% of the increase in aggregate French exports between 1994 and 2006 is due to firms that didn't export before 1994 (new exporters). Restricting attention to Born Global firms, they account for around one third, or 25.5% of the increase in aggregate French exports.²⁵ Together with MNCs, they cumulate a staggering 58.2% of French aggregate exports over the period.²⁶ These figures suggest that BGs do also play an important role in French exports, as in Chile.

4.1.2 Number of countries and products

Bellone et al. (2023) show that, on average, BGs in the French manufacturing sector export more products to more countries. In this subsection, we complement these findings by examining how French BGs' product and geographical scope vary with exporting age.

We restrict the analysis to BG new exporters with a spell duration of five consecutive years or more (except for the survival rate analysis), mostly for comparison with the results of the previous section for new exporter MNCs.²⁷

Figure 5 displays the number of countries (left panel) and products (right panel) for BGs (solid line) relative to old experienced French exporters (dashed line). Besides serving more destinations and products than experienced exporters from the onset, they also appear to expand faster along those two dimensions of their extensive margins. Looking at Figure 1 for MNCs,

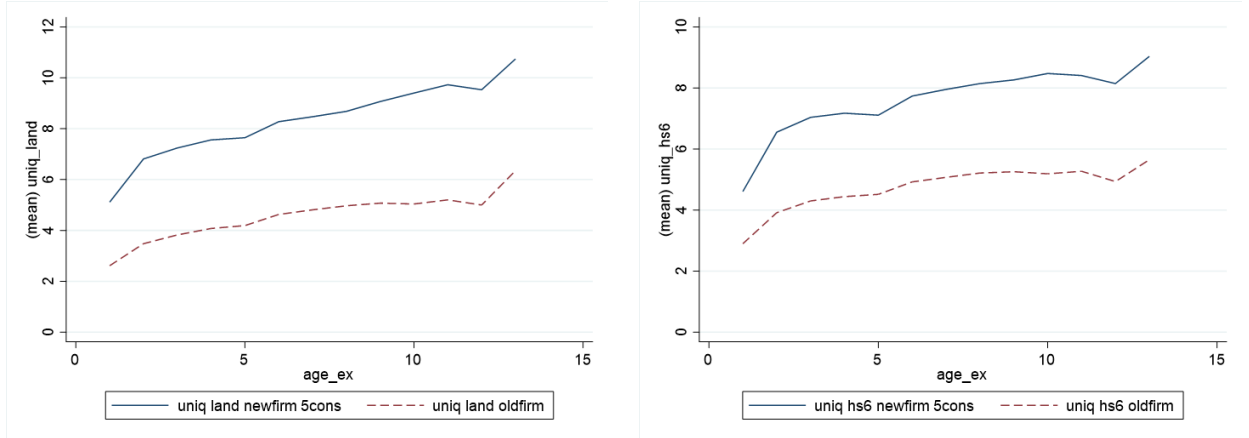
²⁵In our sample, exports increased from EUR 119bn in 1994 to EUR 225bn in 2006. Among new exporters they rose from EUR 2.07bn to EUR 80.8bn. Among the subset of BG new exporters, exports rose from 1.27bn to 28.3bn.

²⁶In our sample, the total export value (euros) in year 1994 is 1.19×10^{11} , with old exporters accounting for 1.17×10^{11} and new exporters for 2.07×10^9 . Among the new exporters in 1994, the export value of ever MNCs or BGs is 1.68×10^9 and the export value of BGs only is 1.27×10^9 . The total export value in 2006 is 2.25×10^{11} with old exporters accounting for 1.44×10^{11} and new exporters for 8.08×10^{10} . Among the new exporters in 2006, the export value of ever MNCs or BGs is 6.34×10^{10} and the export value of BGs only is 2.83×10^{10} . The growth contribution of new exporters is calculated as $\frac{8.08 \times 10^{10} - 2.07 \times 10^9}{2.25 \times 10^{11} - 1.19 \times 10^{11}}$ and the growth contribution of ever MNCs or BGs is calculated $\frac{6.34 \times 10^{10} - 1.68 \times 10^9}{2.25 \times 10^{11} - 1.19 \times 10^{11}}$. The growth contribution of BGs only is calculated $\frac{2.83 \times 10^{10} - 1.27 \times 10^9}{2.25 \times 10^{11} - 1.19 \times 10^{11}}$.

²⁷Appendix D.1 shows results in the whole sample.

BGs appear to serve relatively fewer destinations and products, while expanding more gradually along both dimensions. Therefore, the expansion along both product and destination dimensions of French BGs resembles more the pattern observed for French MNCs than the one observed for old French exporters.

Figure 5: Number of exports destinations (left) and export products (right), by age and new firm status



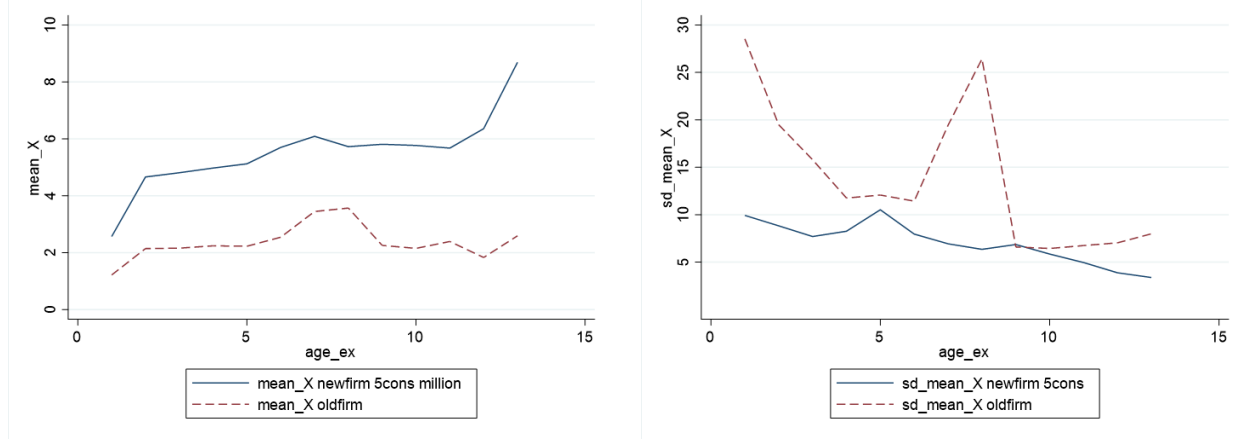
Notes: The sample covers the period from 1994 to 2006 and includes new exporters active in 5 consecutive years (or more). Destinations are defined at the country level; products are defined at the HS6 level. BG new exporters are represented by a solid blue curve, while non-BG new exporters are captured by a dashed red curve.

4.1.3 Export values: level and dispersion

Besides gradually expanding at their extensive margins, do new entities (BGs) also start small, gradually increasing export volumes? Figure 6, left panel, displays an ever increasing mean export value at the firm level by exporting age for new entities (BGs, solid blue line), which is not apparent among existing French firms (dashed red line). Although BGs are also large exporters (albeit of smaller size than MNCs), the similar left panel in Figure 2 shows that the mean value of exports for MNCs 'plateaus' after some years.

To further inquire whether the gradual expansion of French BGs is related to sales uncertainty in newly entered markets, the right panel of Figure 6 reports the normalized standard deviation of firm export values by exporting age and new entity status. For both groups of new exporters, uncertainty falls with exporting age, albeit at a gradual slower pace amongst BGs. In addition, BGs (blue solid line) experience markedly lower uncertainty at all exporting ages. Compared to MNCs (Figure 2 right-hand panel), BGs' normalized standard deviation is slightly higher at early ages, decreasing also gradually with exporting tenure.

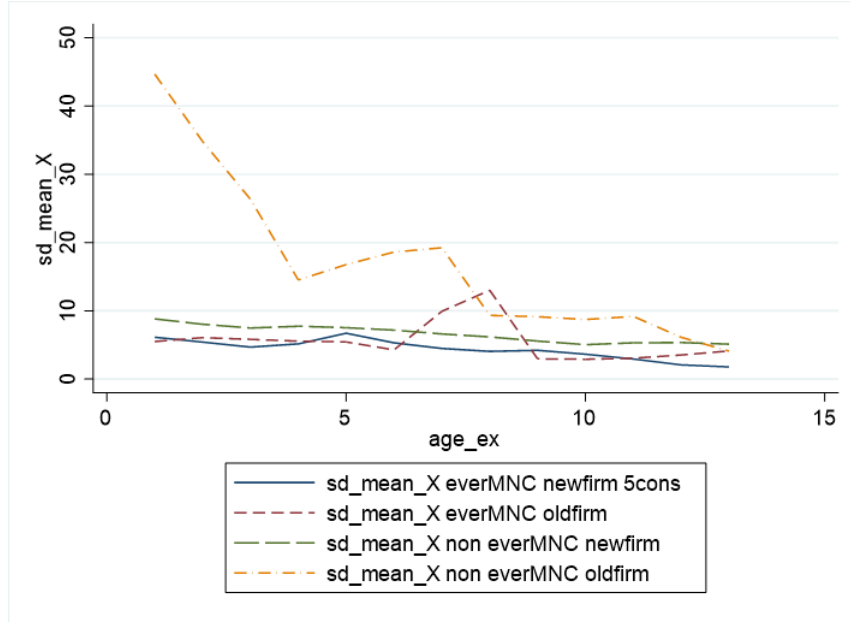
Figure 6: Export value by age and BG status: mean (left panel) and normalized standard deviation (right panel)



Notes: The sample covers the period from 1994 to 2006 and includes new exporters active for 5 consecutive years (or more). Export values are expressed in millions of euros at the firm level. The normalized standard deviation is defined as the ratio of the standard deviation to the mean. The standard deviation is computed across firms by exporting age and BG status. The plot for BG new exporters is captured by a solid blue curve, and the plot for non-BG new exporters is depicted by a dashed red curve.

Because there are MNCs among BGs and existing entities/old firms, and MNCs are also larger exporters consecutively selling abroad for 5 years, we consider four exclusive categories to see which type of exporter drives the observed change in the normalized standard deviation by exporting age in Figure 6. Figure 7 plots the four categories of firms by MNC and BG status, as reported in Table 2 above. Existing firms that are not MNCs (orange dot-dashed line) face the highest initial uncertainty in export values, followed by new firms that are not MNCs (green long-dashed line). For both, the normalized standard deviation of mean export values markedly decreases with exporting age. In contrast, new (solid blue line) and existing (red short-dashed line) firms that are MNCs appear to face much lower initial uncertainty, which also decreases (albeit mildly) by exporting age.

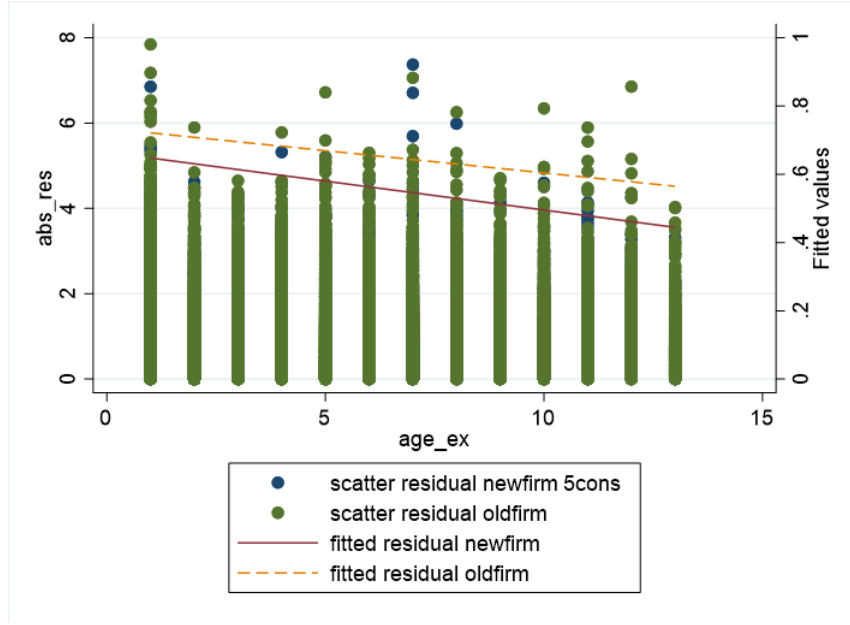
Figure 7: MNCs $\times$ BGs normalized standard deviation



Notes: The sample covers the period from 1994 to 2006 and includes new exporters active for 5 consecutive years (or more). Export values are expressed in millions of euros at the firm level. The normalized standard deviation is defined as the ratio of the standard deviation to the mean. The standard deviation is computed across firms by exporting age, MNC status, and BG status. Among new exporters, the plots for existing non-MNC firms, new non-MNC firms, existing MNCs, and new MNCs are depicted by the orange dot-dashed line, the green long-dashed line, the red short-dashed line, and the solid blue line, respectively.

Just as with MNCs, new entities (BGs) appear to sell more and face less uncertainty than existing entities. But both existing and new entities enter with low export values, rapidly shooting exports while facing decreasing uncertainty as they become more experienced. To discard different growth trends between new and existing entities, Figure 8 plots the absolute values of the estimated residuals along with a line of best fit from estimating equation (1) by BG status. The pattern closely resembles that of MNCs, displayed in the right panel of Figure 3, suggesting that there are similar (parallel) growth trends across the two groups. However, new entities (solid red line) appear to deviate less from trend than existing/old firms (yellow dashed line) do, possibly indicating a lower initial level of uncertainty. Still, that lower level of initial uncertainty for new entities progressively dissipates with exporting age, and does so at a similar speed as it dissipates for existing entities.

Figure 8: Residuals and fitted trend by new/old entity status

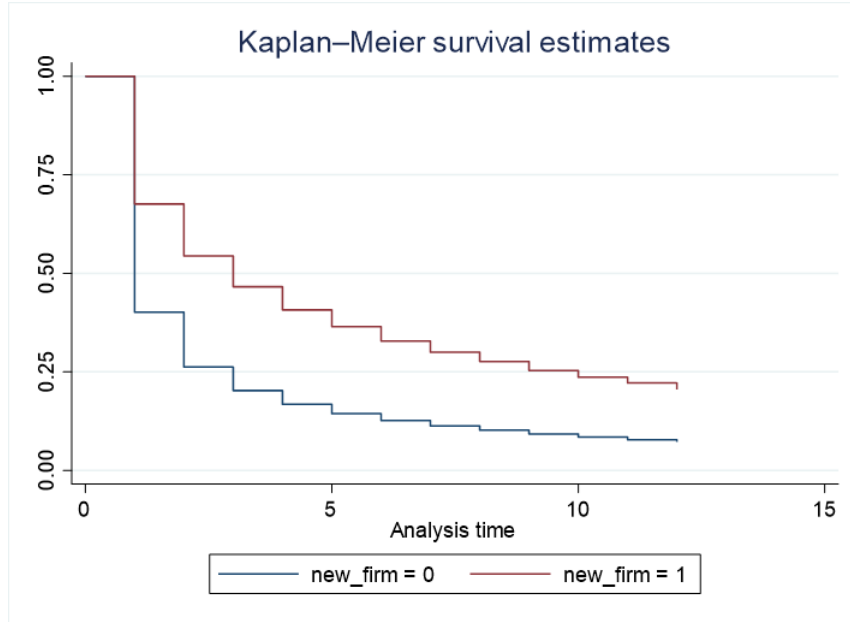


Notes: The sample covers the period from 1994 to 2006 and includes new exporters active for 5 consecutive years (or more). The residuals are plotted in absolute value (left axis), based on a regression of the logarithm of export value on exporting age at the firm level, with firm-specific slopes and firm fixed effects. The right axis shows the trend fitted line values. We distinguish between BG and non-BG status residuals: the fitted trend line for BGs is depicted as a solid red line while the fitted trend line for non-BGs is a dashed yellow line instead.

4.1.4 Survival rate

If new entities (BGs) face less uncertainty, they could also exhibit a higher survival rate compared to other existing new exporters (just as MNCs do). And indeed, figure 9 confirms that BGs are markedly less likely to exit immediately after entering a new spell than existing firms. Yet, their survival rates regularly adjust by exporting age, indicating that some gradual adaptation is reducing the hazard rates for both groups.

Figure 9: Kaplan-Meier survival rate estimates by new entity status and firm exporting age:



Notes: The sample covers the period from 1994 to 2006. Exit is a binary variable that takes value 1 if exports are positive in year t and zero in year $t+1$. We use the *sts* command in Stata to produce the graph. The survival curve for BGs is solid red, and the survival curve for non-BGs is solid blue.

All in all, the reported descriptive evidence points towards French newly created entities/firms not behaving as old/existing exporters, unlike in Chile. Among the subpopulation of new exporters that have exported for five consecutive years, new entities/firms are larger from the onset, face less and decreasing Foreign sales uncertainty by exporting age (Fig. 6), being almost twice more likely to survive right after entering (Fig. 9), serving more destinations, and with more products than existing firms (Fig. 5). As such, new exporting entities/firms (BGs) behave more like new exporting MNCs, appearing slightly smaller and facing higher uncertainty, whilst growing sales for longer (Figs. 6 and 2).

4.2 Regression Results

To better understand whether BGs also expand gradually, in this section we examine differences in export growth, exit and (conditional on surviving) entry, between new (BGs) and existing (new exporter) entities. We examine correspondingly similar econometric specifications as in the previous section for MNCs. Therefore, we replace the MNC_{it} indicator for MNC status of firm i by the time-invariant born-global new entity status indicator BG_i in all equations. The

definition of the three dependent variables is also the same for comparison purposes.

4.2.1 Within-spell dynamics

We first look at within-spell export dynamics, namely growth and exit of existing and new entities by spell age. Recall that a spell is defined at the firm-country-product ijp level. New and existing entities continuously add new/existing products into existing/new countries (or both), enabling us to examine the expansion pattern both geographically and of their product scopes as they mature as exporters.

4.2.1.1 Growth regressions. Table 8 presents the within-spell growth results from estimating two specifications: one with only the BG_i dummy to capture differences in average growth rates by new entity status, and specification (2) with a new entity status dummy, BG_i , fully interacted with spell age, to capture differences in growth patterns between new (BGs) and existing entities.²⁸ The reported results indicate that new French entities significantly grow initially, to reach their steady states by spell age 6, just as existing entities do (column 1). Relative to MNCs however, born-global firms show a significantly higher within-spell growth rate at ages 3 and 5. This suggests that BGs grow slightly faster than other new exporters, including MNCs.

²⁸Contrary to what happens in Table 3 for MNCs, there is no BG dummy at the end of column 1 because it is absorbed by the spell fixed effect.

Table 8: BGs Growth Regressions Within Spell

	(1) Growth		(2) Growth	
	Coef.	SE	Coef.	SE
age2	0.638***	(0.038)	0.605***	(0.040)
age3	0.150***	(0.032)	0.114***	(0.034)
age4	0.079***	(0.027)	0.069**	(0.030)
age5	0.052**	(0.023)	0.026	(0.026)
age6	0.036**	(0.018)	0.022	(0.020)
age7	0.013	(0.014)	0.016	(0.017)
BG_age2			0.062*	(0.033)
BG_age3			0.069***	(0.026)
BG_age4			0.020	(0.025)
BG_age5			0.050*	(0.025)
BG_age6			0.027	(0.021)
BG_age7			-0.008	(0.021)
BG_age8			0.000	(.)
_cons	0.223**	(0.103)	0.222**	(0.102)
gravity	yes		yes	
firm level	yes		yes	
year FE	yes		yes	
spell FE	yes		yes	
<i>N</i>	685,190		685,190	
<i>R</i> ²	0.201		0.201	

Notes: The sample covers the period 1994–2006. We use a linear regression. The dependent variable is export growth, as defined in Appendix A.2.2. Growth rates are computed from age 2. The constant term captures growth at age 8 or older, while the age dummies capture deviations from that baseline. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * $p < 0.10$, ** $p < .05$, *** $p < .01$

4.2.1.2 Exit regressions. Table 9 reports the results of estimating the specification (3) with fully interacted spell age and BG_i dummy, under columns 2 and 4 for different distributions of random effects.²⁹ Instead, the results of estimating a similar hazard specification but without spell age interactions and by new entity status, are reported under columns 1 and 3. The results are very similar to those of MNCs. BGs have a lower exit rate at all spell ages than existing new exporters, since the age interacted terms appear all significantly below 1. However, MNCs have lower hazard rates than BGs, with the estimated coefficients of the interaction terms being generally smaller (and decrease as spell age increases) for MNCs.

Therefore, global-born French new exporters appear to behave much like other new exporters, including MNCs, starting small in their export spells, with high initial exit probabilities, but quickly growing upon surviving, reaching their steady-states after 5-6 years. Yet, in terms of

²⁹Recall that all coefficients are reported in exponential form, where a coefficient above 1 implies a positive original value, and a coefficient below 1 implies a negative original value.

both larger initial sales and lower exit probabilities upon spell entry, they resemble more MNC new exporters, than non-MNC ones. We next look at whether the rationale behind the detected behavior could be attributed to learning about profitability that happens at the firm level, and which would hence spill over across export spells.

Table 9: BGs Exit Regressions Within Spell

	(1) Exit	(2) Exit	(3) Exit	(4) Exit
age1	14.776 (0.696)***	14.816 (0.698)***	16.308 (0.774)***	16.348 (0.777)***
age2	6.706 (0.321)***	6.778 (0.326)***	7.380 (0.356)***	7.453 (0.362)***
age3	5.553 (0.267)***	5.585 (0.270)***	6.118 (0.297)***	6.152 (0.301)***
age4	5.019 (0.242)***	5.005 (0.244)***	5.534 (0.269)***	5.522 (0.272)***
age5	4.595 (0.222)***	4.628 (0.227)***	5.070 (0.248)***	5.112 (0.253)***
age6	4.476 (0.218)***	4.410 (0.218)***	4.942 (0.243)***	4.874 (0.243)***
age7	4.196 (0.205)***	4.183 (0.209)***	4.633 (0.229)***	4.625 (0.233)***
age8	3.986 (0.194)***	3.904 (0.192)***	4.402 (0.216)***	4.315 (0.214)***
BG_age1		0.925 (0.004)***		0.926 (0.004)***
BG_age2		0.911 (0.008)***		0.913 (0.008)***
BG_age3		0.920 (0.010)***		0.921 (0.009)***
BG_age4		0.937 (0.011)***		0.936 (0.011)***
BG_age5		0.918 (0.013)***		0.916 (0.013)***
BG_age6		0.961 (0.016)**		0.959 (0.016)**
BG_age7		0.937 (0.018)***		0.935 (0.018)***
BG_age8		0.975 (0.016)		0.973 (0.016)*
BG	0.926 (0.003)***		0.927 (0.003)***	
gravity	yes	yes	yes	yes
firm level	yes	yes	yes	yes
year FE	yes	yes	yes	yes
spell random	normal	normal	gamma	gamma
N	1495998	1495998	1495998	1495998

Notes: The sample covers the period 1994–2006. The year 2006 is used to identify left-censored spells. The dependent variable is exit hazard. See Appendix A.2.3 for a detailed definition of exit. We estimate discrete-time proportional hazards models with random effects. For normal and gamma-distributed random effects, we use `xtcloglog` and `pgmhaz8`, respectively. “Age 8” refers to age 8 or older. See Appendix A.2.5 for the list and definitions of gravity and firm-level variables. Standard errors are in parentheses. * p<0.10, ** p<.05, *** p<.01

4.2.2 Dynamic complementarities across markets

As in the corresponding section for MNCs, here we consider potential dynamic complementarities between markets, defined at the country-product level by new entity status (BGs). And just as in the previous subsection, we estimate identical corresponding specifications as for MNCs, replacing the MNC_{it} dummy by the BG_i one.

4.2.2.1 Growth regressions. The results reported in Table 10 obtain from estimating specification (4) and for similarly ordered combinations of controls and fixed effects (FEs). Under columns 2 and 4, the estimated coefficient of BG_i is missing because BG_i is firm-specific and time-invariant and is thus absorbed by firm FEs. The results are very similar to those of MNCs: Because the interaction term α_{YMPbg} is also not significant, differential growth patterns in spell age are not present for BGs neither, growing at similar rates as other new exporters, MNCs included. But the results reported in the last row of the table are consistent with marked age effects in the growth of BG new exporters controlling for size and TFP growth.

Table 10: BGs Growth Regressions Baseline

	(1) Growth	(2) Growth	(3) Growth	(4) Growth
α_{YMP}	0.173 (0.023)***	0.225 (0.024)***	0.175 (0.031)***	0.252 (0.035)***
BG	-0.003 (0.008)	0.000 (.)	0.008 (0.014)	0.000 (.)
α_{YMPbg}			-0.002 (0.047)	-0.052 (0.048)
gravity	yes	yes	yes	yes
firm level	yes	yes	yes	yes
year FE	yes	yes	yes	yes
country FE	yes	yes	yes	yes
product FE	yes		yes	
firm FE		yes		yes
N	834,134	834,134	834,134	834,134
R^2	0.020	0.058	0.020	0.058
$\alpha_{YMP} + \alpha_{YMPbg}$			0.174***	0.200***

Notes: The sample covers the period from 1994 to 2006. The OLS dependent variable is export growth, defined in Appendix A.2.2. Growth rates are computed from age 2. In specifications (1), and (3), we use product fixed effects, and in specifications (2), and (4), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables. * $p < 0.10$, ** $p < .05$, *** $p < .01$

4.2.2.2 Exit regressions. Turning now to exit rates early rather than later by spell age, Table 11 reports the results of estimating the corresponding specification (5) for new entities (BGs). The results are also very similar to those of MNCs. Under column 3, the estimated coefficient β_{YMPbg} is statistically and significantly negative, indicating that BGs exit less frequently than other new exporters. However, the sum of coefficients $\beta_{YMPbg} + \beta_{YMP}$ is significantly positive, which is not the case for MNCs. Hence, and much like other new exporters, BGs also exhibit a significantly higher exit rate in their very first spell, i.e. first-year-first-market-first-product, relative to the average of later spells.

Table 11: BGs Exit Regressions Baseline

	(1) Exit	(2) Exit	(3) Exit	(4) Exit
β_{YMP}	0.137 (0.010)***	-0.036 (0.006)***	0.128 (0.011)***	-0.054 (0.007)***
BG	-0.015 (0.004)***	0.000 (.)	-0.011 (0.008)	0.000 (.)
β_{YMPbg}			-0.064 (0.023)***	0.036 (0.019)*
gravity	yes	yes	yes	yes
firm level	yes	yes	yes	yes
year FE	yes	yes	yes	yes
country FE	yes	yes	yes	yes
product FE	yes		yes	
firm FE		yes		yes
N	1,465,731	1,443,322	1,465,731	1,443,322
R^2	0.228	0.309	0.228	0.310
$\beta_{YMP} + \beta_{YMPbg}$			0.063***	-0.018

Notes: The sample covers the period from 1994 to 2006. The OLS dependent variable is export exit, defined in Appendix A.2.3. Year 2006 identifies the exit in 2005. In specifications (1), and (3), we use product fixed effects, and in specifications (2), and (4), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables. * p<0.10, ** p<.05, *** p<.01

4.2.2.3 Conditional Entry regressions The results reported in Table 12 are very similar to those of MNCs, reported in Table 7 out of estimating specification (6). The absolute values of γ_{YMBg} and γ_{YPbg} are slightly smaller than those of γ_{YMmnc} and γ_{YPmnc} , suggesting that BGs are relatively more active in adding products to the same initial country and less active in exporting the same initial product to other countries than MNCs. But just as for MNCs, French BG new exporters display significant age effects in their geographical and product scope expansions, also adding early more new products than new destinations.

Table 12: Entry Regressions of BGs

	(1)	(2)	(3)	(4)
	Entry	Entry	Entry	Entry
γ_Y	0.00081 (0.00010)***	-0.00008 (0.00011)	0.00102 (0.00013)***	0.00001 (0.00013)
γ_{YM}	0.02566 (0.00170)***	0.02550 (0.00154)***	0.02964 (0.00291)***	0.02883 (0.00261)***
γ_{YP}	0.00121 (0.00022)***	0.00121 (0.00020)***	0.00075 (0.00027)***	0.00077 (0.00025)***
γ_{Ybg}			-0.00053 (0.00020)***	-0.00021 (0.00020)
γ_{YMBg}			-0.00799 (0.00348)**	-0.00669 (0.00316)**
γ_{YPBg}			0.00116 (0.00045)***	0.00115 (0.00042)***
BG	0.00084 (0.00009)***	0.00000 (.)	0.00047 (0.00008)***	0.00000 (.)
gravity	yes	yes	yes	yes
firm level	yes	yes	yes	yes
year FE	yes	yes	yes	yes
country FE	yes	yes	yes	yes
product FE	yes		yes	
firm FE		yes		yes
N	49,217,228	49,217,228	49,217,228	49,217,228
R^2	0.01125	0.01348	0.01130	0.01352
$\gamma_Y + \gamma_{Ybg}$			0.00049***	-0.00020
$\gamma_{YM} + \gamma_{YMBg}$			0.02166***	0.02214***
$\gamma_{YP} + \gamma_{YPBg}$			0.00191***	0.00192***

Notes: The sample covers the period from 1994 to 2006. The OLS dependent variable is export entry, defined in Appendix A.2.4 . In specifications (1), and (3), we deploy product fixed effects, and in specifications (2), and (4), we instead estimate with firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables. * p<0.10, ** p<.05, *** p<.01

Overall, the results reported in this section confirm that French BGs (new entities) display exporting age dynamics that are (i) similar to existing entities (old exporters) and MNCs along the intensive growth- (Tab.10) and conditional entry extensive (Tab.12) margins, starting small but quickly expanding upon surviving. When it comes to (ii) exit, BGs behave more like existing non-MNC new exporters, exiting earlier rather than later, albeit at a smaller rate than other new exporters do (Tab.11, column 3). Recall that we did not find an age pattern in MNCs exit decisions out of exporting. BGs thus appear as an intermediate case between non-BG-non-MNC new exporters, and new exporters that are MNCs.

5 Conclusions

In this paper, we examined the dynamics of the universe of French exporting firms between 1993 and 2006, using highly disaggregated customs and balance sheet data. We found surprisingly small differences in the exporting dynamics of new multinational and born-global exporters relative to smaller new exporters.

Although the literature has established that MNCs and BG firms are different from the median exporter, here we have found that (1) MNC and BG new exporters display growth and exit rates that fall substantially with spell age, just as for smaller new exporters, even when controlling for size, firm-level TFP growth and unobserved (spell) heterogeneity. (2) Successful MNC and BG new exporters tend to have faster growth at the beginning of their first spell relative to later spells, just as smaller non-BG or non-MNC new exporters. In terms of cross-spell exit patterns, BGs exhibit higher exit in their first spell just like other new exporters while MNCs don't. (3) Conditional on surviving, they also tend to enter more destinations and add products earlier rather than later. We believe that these age dependent patterns reflect gradual and complementary /interdependent decisions to enter foreign markets that render third-country trade policy effects normal, rather than the exception. But its exploration is left for future research.

It has been argued that MNCs and BGs experience a different internationalization process, thanks to superior knowledge and assets. Blum et al. (2020) show indeed that these exporters, while being few, account for a large share of new Chilean exporters' growth. They also suggest that these firms behave more like experienced exporters. Our results suggest that these findings may not extend to other countries. In our French sample, MNC and BG new exporters look more similar to other new exporters than to older MNC and BG exporters, respectively. Our results suggest that theories based on learning, customer base accumulation, marketing investment or capability-building may apply to MNC and BG exporters too. Furthermore, the mechanism behind age dependence applies to within-spell dynamics as well as cross-spell dynamics, with the exception of exit in the case of MNCs. This suggests that positive dynamic complementarities in entry into different product-country markets are at work, even among MNCs and BGs.

A limitation of our exercise is that it does not capture the partial or general equilibrium impli-

cations of firms’ export dynamics process. An interesting avenue for further research would be to examine aggregate export dynamics using a structural model of firm export decisions that features both age dependence and dynamic complementarities in individual export dynamics.

References

- Aeberhardt, Romain, Ines Buono, and Harald Fadinger**, “Learning, incomplete contracts and export dynamics: Theory and evidence from French firms,” *European Economic Review*, 2014, 68, 219–249.
- Albornoz, Facundo, Héctor F Calvo-Pardo, Gregory Corcos, and Emanuel Ornelas**, “Sequential Exporting,” *Journal of International Economics*, 2012, 88 (1), 17–31.
- , **Héctor F Calvo Pardo, Gregory Corcos, and Emanuel Ornelas**, “Sequentially exporting products across countries,” *Journal of International Economics*, 2023, 142, 103735.
- Alessandria, George A., Robert C Johnson, and Kei-Mu Yi**, “Perspectives on Trade and Structural Transformation,” NBER Working Paper 28720 2021.
- Alessandria, George, Costas Arkolakis, and Kim J. Ruhl**, “Firm Dynamics and Trade,” *Annual Review of Economics*, 2021, 13 (1), 253–280.
- Antràs, Pol and Stephen R. Yeaple**, “Multinational Firms and the Structure of International Trade,” in G. Gopinath, . Helpman, and K. Rogoff, eds., *Handbook of International Economics*, Vol. 4 of *Handbook of International Economics*, Elsevier, 2014, chapter 0, pp. 55–130.
- Araujo, Luis, Giordano Mion, and Emanuel Ornelas**, “Institutions and export dynamics,” *Journal of International Economics*, 2016, 98, 2–20.
- Arkolakis, Costas, Theodore Papageorgiou, and Olga Timoshenko**, “Firm Learning and Growth,” *Review of Economic Dynamics*, 2018, 27, 146–168.
- Bellone, Flora, Catherine Laffineur, and Sophie Pommet**, “Public Aid and the Performance of Born Globals,” *Economie et Statistique/Economics and Statistics*, 2023, 540, 43–60.

- Berman, Nicolas, Vincent Rebeyrol, and Vincent Vicard**, “Demand Learning and Firm Dynamics: Evidence from Exporters,” *The Review of Economics and Statistics*, 2019, *101* (1), 91–106.
- Bernard, Andrew B, J. B. Jensen, Stephen J Redding, and Peter K Schott**, “Global Firms,” *Journal of Economic Literature*, 2018, *56* (2), 565–619.
- Berthou, Antoine and Vincent Vicard**, “Firms’ export dynamics: experience versus size,” *The World Economy*, 2015, *38* (7), 1130–1158.
- Beveren, Ilke Van, Andrew B Bernard, and Hylke Vandenbussche**, “Concording EU trade and production data over time,” Technical Report, National Bureau of Economic Research 2012.
- Blum, Bernardo S., Sebastian Claro, Ignatius Horstmann, and Trevor Tombe**, “The DNA of New Exporters: Spin-offs and FDI at the Extensive Margin of Trade,” *American Economic Review: Insights*, September 2020, *2* (3), 397–408.
- Cadot, Olivier, Leonardo Iacovone, Martha Denisse Pierola, and Ferdinand Rauch**, “Success and failure of African exporters,” *Journal of Development Economics*, 2013, *101*, 284–296.
- Carrère, Céline and Vanessa Strauss-Kahn**, “Export Survival and the Dynamics of Experience,” *Review of World Economics (Weltwirtschaftliches Archiv)*, 2017, *153* (2), 271–300.
- Cebreros, Alfonso**, “The Rewards of Self-Discovery: Learning and Firm Exporter Dynamics,” Working Paper 2016-08, Banco de México 2016.
- Chen, Cheng, Chang Sun, and Hongyong Zhang**, “Learning and Information Transmission Within Multinational Corporations,” *European Economic Review*, 2022, *143*, 104016.
- , **Tatsuro Senga, Chang Sun, and Hongyong Zhang**, “Uncertainty, imperfect information, and expectation formation over the firm’s life cycle,” *Journal of Monetary Economics*, 2023, *140*, 60–77.
- Choquette, Eliane, Morten Rask, Davide Sala, and Philipp Schröder**, “Born Globals—Is there fire behind the smoke?,” *International Business Review*, 2017, *26* (3), 448–460.

- Conconi, Paola, André Sapir, and Maurizio Zanardi**, “The Internationalization Process of Firms: From Exports to FDI,” *Journal of International Economics*, 2016, 99, 16–30.
- , **André Sapir, and Maurizio Zanardi**, “The internationalization process of firms: From exports to FDI,” *Journal of International Economics*, 2016, 99 (C), 16–30.
- Eaton, Jonathan, Marcela Eslava, Maurice Kugler, and James Tybout**, “Export Dynamics in Colombia: Firm Level Evidence,” in Elhanan Helpman, Dalia Marin, and Thierry Verdier, eds., *The Organization of Firms in a Global Economy*, Cambridge, MA: Harvard University Press, 2008, chapter 8.
- Esteve-Pérez, Silviano**, “Previous experience, experimentation and export survival: Evidence from firm-product-destination level data,” *The World Economy*, 2021, 44 (9), 2638–2682.
- Fillat, José L. and Stefania Garetto**, “Risk, Returns, and Multinational Production *,” *The Quarterly Journal of Economics*, 09 2015, 130 (4), 2027–2073.
- Fitzgerald, Doireann, Stefanie Haller, and Yaniv Yedid-Levi**, “How Exporters Grow,” NBER Working Paper 21935 2019.
- Freund, Caroline and Martha Denisse Pierola**, “Export Entrepreneurs: Evidence from Peru,” Policy Research Working Paper Series 5407, The World Bank 2010.
- **and** —, “Export Superstars,” *The Review of Economics and Statistics*, 2015, 97 (5), 1023–32.
- **and** —, “The Origin and Dynamics of Export Superstars,” *The World Bank Economic Review*, 2020, 34 (1), 28–47.
- Gumpert, Anna, Haishi Li, Andreas Moxnes, Natalia Ramondo, and Felix Tintelnot**, “The life-cycle dynamics of exporters and multinational firms,” *Journal of International Economics*, 2020, 126, 103343.
- Inagaki, Kazuma**, “Born Global, New Exporter Dynamics, and the Aggregate Gains from Trade,” *Available at SSRN*, September 2025, p. 1–57.
- Johanson, Jan and Jan-Erik Vahlne**, “The Internationalization Process of the Firm—A Model of Knowledge Development and Increasing Foreign Market Commitments,” *Journal of International Business Studies*, March 1977, 8 (1), 23–32.

- Knight, Gary A and S Tamar Cavusgil**, “Innovation, organizational capabilities, and the born-global firm,” *Journal of International Business Studies*, March 2004, *35* (2), 124–141.
- Lawless, Martina and Zuzanna Studnicka**, “Old Firms and New Export Flows: Does Experience Increase Survival?,” Working Papers 19/19, School of Economics, University College Dublin 2019.
- Levinsohn, James and Amil Petrin**, “Estimating Production Functions Using Inputs to Control for Unobservables,” *The Review of Economic Studies*, 2003, *70* (2), 317–341.
- Li, Shengyu**, “A Structural Model of Productivity, Uncertain Demand, and Export Dynamics,” *Journal of International Economics*, 2018, *115*, 1–15.
- Nguyen, Daniel X.**, “Demand uncertainty: Exporting delays and Exporting Failures,” *Journal of International Economics*, 2012, *86* (2), 336–344.
- Pierce, Justin R. and Peter K. Schott**, “Concording U.S. Harmonized System Categories over Time,” Finance and Economics Discussion Series 2012-16, Board of Governors of the Federal Reserve System (U.S.) 2012.
- Piveteau, Paul**, “An Empirical Dynamic Model of Trade with Consumer Accumulation,” *American Economic Journal: Microeconomics*, November 2021, *13* (4), 23–63.
- Ramondo, Natalia, Veronica Rappoport, and Kim J. Ruhl**, “The Proximity-Concentration Tradeoff under Uncertainty,” *The Review of Economic Studies*, 08 2013, *80* (4), 1582–1621.
- Ruhl, Kim J. and Jonathan L. Willis**, “New Exporter Dynamics,” *International Economic Review*, 2017, *58*, 703–726.
- Timoshenko, Olga A.**, “Learning versus Sunk Costs Explanations of Export Persistence,” *European Economic Review*, 2015, *79*, 113–128.

Appendix [not for publication]

New Exporter Dynamics of Born-Global and Multinational Firms

This appendix provides additional proofs, robustness checks, and supplementary material referenced in the main text.

A Appendix: Data

A.1 Data sources

The French customs data (DGDDI) include all export transactions from 1993 to 2006, categorized by firm, HS6 product, destination country, and year. We apply the HS6 concordance table in Van Beveren et al. (2012), which is based on the method developed in Pierce and Schott (2012), to address revisions in the HS classification. Values are reported in euros. We drop countries that account for less than 5% of French exports in order to reduce the size of our dataset. We merge the customs data with the CEPII Gravity dataset to incorporate information on destination-country characteristics. The resulting dataset contains export values for 403 657 firms, 4212 HS6 products, 90 countries, and 14 years.

We use balance-sheet data from the French tax authority (FICUS) and the financial linkages data (LIFI) to provide firm-level characteristics. The FICUS dataset provides information on a firm's primary industry, total sales, employment, capital stock, cost of materials, total debts, total net assets, and more. The LIFI dataset provides information on whether a French firm is owned by a foreign company and whether it has foreign affiliates.

Our dataset merges these four sources, excluding firms in the distribution sectors, namely the two-digit NACE sectors 50, 51, and 52. We also exclude firms that have ever been owned by a company located in the FSF–IMF (2000) tax-haven list.

A.2 Definition of variables

A.2.1 Exporting age

Exporting age is defined as the difference between the current year and the year a firm first started exporting, plus one. Note that we define age as time elapsed since the firm's export start. If a firm began exporting in 1999, stopped in 2001, and then restarted in 2003, its exporting age would range from 1 to 3 from 1999 to 2001 and then resume at 5 in 2003.

A.2.2 Export growth

Denote by x_{ijpt} the recorded sales of French firm i in country j of HS6 product p in year t . $Growth_{ijpt}$ is the annual growth rate of firm-country-product exports, measured in FOB value, defined as:

$$Growth_{ijpt} = \ln(x_{ijpt}) - \ln(x_{ijpt-1}).$$

A.2.3 Export exit

$Exit_{ijpt}$ is a binary variable that is defined when the firm-country-product has positive exports in year t . It takes the value 1 when exports for that triplet become 0 in year $t + 1$ and remain 0 thereafter, and 0 otherwise.³⁰ Therefore, in this setting, the definition of exit corresponds to a permanent exit.

$$Exit_{ijpt} = \begin{cases} 1 & \text{if } x_{ijpt} > 0 \text{ and } x_{ijp,t+i} = 0 \text{ for all } i \geq 1 \\ 0 & \text{if } x_{ijpt} > 0 \text{ and } x_{ijp,t+i} > 0 \text{ for some } i \geq 1. \end{cases}$$

A.2.4 Export entry

$Entry_{ijpt}$ is a binary variable that is defined for firm-country-product triplets with no exports in year $t - 1$. It takes value one when the triplet has positive exports in year t and takes value 0 otherwise:

$$Entry_{ijpt} = \begin{cases} 1 & \text{if } x_{ijpt} > 0 \text{ and } x_{ijpt-1} = 0 \\ 0 & \text{if } x_{ijpt} = 0 \text{ and } x_{ijpt-1} = 0. \end{cases}$$

Our definition of non-entry has implications for the construction of the sample. To capture non-entry, we must expand our native dataset to include firm-product-country-year observations that are never observed in the customs data. More precisely, for all observed firm-product pairs,

³⁰The year 2006 is used to identify whether export spells are right-censored or failed by the end of 2005.

we span over all possible countries and years. By doing so, we allow for all the possible entry patterns a firm may consider for its products.³¹ Our prediction applied to firms that experienced a first export success. Therefore, we exclude from the sample all firms that experienced an immediate exit from a product-country even if those firms exported again. We also exclude pre-exporting years and the first year of exporting. Finally, we exclude all post-entry observations in the entry analysis, in the sense that firm-product-country triplets with $Entry_{ijpt} = 1$ leave the sample from $t + 1$ onward.

A.2.5 Gravity and firm-level variables

Tab.A.1: Definition of Variables

Variable	Definition
<i>Panel A: Gravity Variables (CEPII)</i>	
<code>contig</code>	Binary: contiguous border with France
<code>comlang_off</code>	Binary: common official or primary language
<code>col45</code>	Binary: colonial relationship post-1945
<code>colony</code>	Binary: ever in colonial relationship
<code>comleg</code>	Binary: common legal origins before transition
<code>comcur</code>	Binary: common currency
<code>gatt_d</code>	Binary: destination is GATT/WTO member
<code>rta</code>	Binary: Regional Trade Agreement (source: WIO)
<code>ACP</code>	Binary: ACP country in current year
<code>distw</code>	Weighted distance to France (population-weighted, km)
<code>pop_d</code>	Destination population (millions)
<code>gdpcap_d</code>	Destination GDP per capita (current US\$)
<i>Panel B: Firm-Level Variables (FICUS & LIFI)</i>	
<code>TFPg</code>	Total factor productivity growth
<code>appgr0</code>	Binary: independent firm (not part of a group)
<code>MNC_it</code>	Binary: firm is an MNC in current year
<code>L1lncatotal</code>	Log total sales lagged one year (millions of euros)
<code>leverage_y2</code>	Binary: firm in 2nd tercile of leverage ratio (leverage = total debt / total net assets)
<code>leverage_y3</code>	Binary: firm in 3rd tercile of leverage ratio

Notes: Gravity variables are drawn from the CEPII Gravity database. Firm-level variables come from the FICUS (balance sheets) and LIFI (financial linkages) datasets. Binary variables take value 1 if the condition holds and 0 otherwise. The leverage ratio is defined as total debt divided by total net assets; tercile dummies are computed within each year.

³¹Implicitly, we assume that the relevant set of products a firm considers for exports is the one it eventually exports. This is the best we can do in the absence of data on domestic sales by product.

Total Factor Productivity (TFP) is estimated at the firm–year level using the Levinsohn–Petrin method (Levinsohn and Petrin, 2003). Within each NACE 1-digit industry subsample, TFP is obtained as the residual from a regression of firm output on capital and labor, using intermediate inputs as a proxy for unobserved productivity.³² We then compute firm-level TFP growth. This variable is included in all estimated models to net out supply-side shocks from export dynamics.

Tab.A.2: Descriptive Statistics

Variable name	Mean	Min	Max	Observations
<i>Country-level variables</i>				
contig	0.358	0	1	2,440,736
comlang_off	0.326	0	1	2,440,736
col45	0.131	0	1	2,440,736
distw	3,029	474	18,894	2,440,736
pop_d	60.17	0.06	1,311	2,440,736
gdpcap_d	23,445	65	72,960	2,403,963
colony	0.201	0	1	2,440,736
gatt_d	0.924	0	1	2,440,736
rta	0.741	0	1	2,440,736
comleg	0.461	0	1	2,440,736
comcur	0.317	0	1	2,440,736
ACP	0.055	0	1	2,440,736
<i>Firm-level variables</i>				
TFPg	0.858	-1.000	10,710	2,061,863
appgr0	0.452	0	1	2,321,413
MNC_it	0.184	0	1	2,440,736
leverage_y2	0.359	0	1	2,103,070
leverage_y3	0.263	0	1	2,103,070
L1lnecatotal	1.949	-6.908	9.401	2,233,198

Notes: Country-level variables come from the CEPII Gravity database. Firm-level variables come from the FICUS and LIFI datasets.

³²The NACE 1-digit industries are Agriculture, Fishing, Mining, Manufacturing, Energy and Water, Construction, Trade, Tourism, Transportation and Communication, Finance, Real Estate and Professional Activities, Government, Education, Health, Other Services, Household Services, and Non-resident Activities. Firm's output equals ventes de marchandises + production vendue de biens et services + production immobilisée + production stockée. Labor force is effectif salarié moyen. Intermediates equal achats de matières premières et autres approvisionnements y compris droits de douane + variation de stocks (matières premières et approvisionnements) + autres achats et charges externes + achats de marchandises y compris droits de douane + variation de stocks (marchandises). Capital equals immobilisations corporelles + immobilisations incorporelles + immobilisations financières - total de l'actif immobilisé (amortissements et provisions).

B Appendix: Blum table using their definition of new exporter

Tab.A.3: Firm-level characteristics by exporter type (Blum et al. definition: new exporter in year t did not export in $t - 1$)

	All Exporters					New Exporters				
	All	Non Age-1 New	Age-1 New	Ever-MNC	Non-MNC	Ever-MNC	Non-MNC	Born Global	Non Born Global	Global
Number of firms	45,651	34,000	11,650	6,266	39,385	717	10,933	1,284	10,099	
Share of exporters	1.000	0.7446	0.2554	0.1374	0.8626	0.0157	0.2397	0.0282	0.2210	
Total exports	1.82×10^8	1.78×10^8	3.52×10^6	1.05×10^8	7.65×10^7	1.74×10^6	1.78×10^6	1.28×10^6	2.20×10^6	
Share of all exports	1.000	0.9805	0.0195	0.5793	0.4207	0.0095	0.0100	0.0075	0.0117	
Exports per firm (median)	65	154	6	1,133	46	16	6	13	6	
Exports per firm (mean)	3,971	5,218	304	16,761	1,938	2,443	162	989	216	
Total sales per firm (median)	1,961	2,685	732	12,419	1,534	5,197	662	394	799	
Total sales per firm (mean)	20,036	25,085	5,191	76,602	11,038	25,476	3,862	5,939	5,145	
Domestic sales per firm (median)	1,630	2,150	692	8,706	1,299	4,888	626	326	766	
Domestic sales per firm (mean)	16,066	19,867	4,887	59,841	9,100	23,033	3,700	4,950	4,928	
Export intensity (median)	0.0498	0.0777	0.0113	0.1243	0.0440	0.0046	0.0118	0.0466	0.0095	
Export intensity (mean)	0.1566	0.1834	0.0775	0.2367	0.1439	0.0739	0.0777	0.1639	0.0659	
Labor (median)	15	20	7	68	12	33	6	3	7	
Capital (median)	184	256	71	2,057	138	1,000	64	32	79	
Value added per worker (median)	43	44	40	52	42	50	40	33	41	

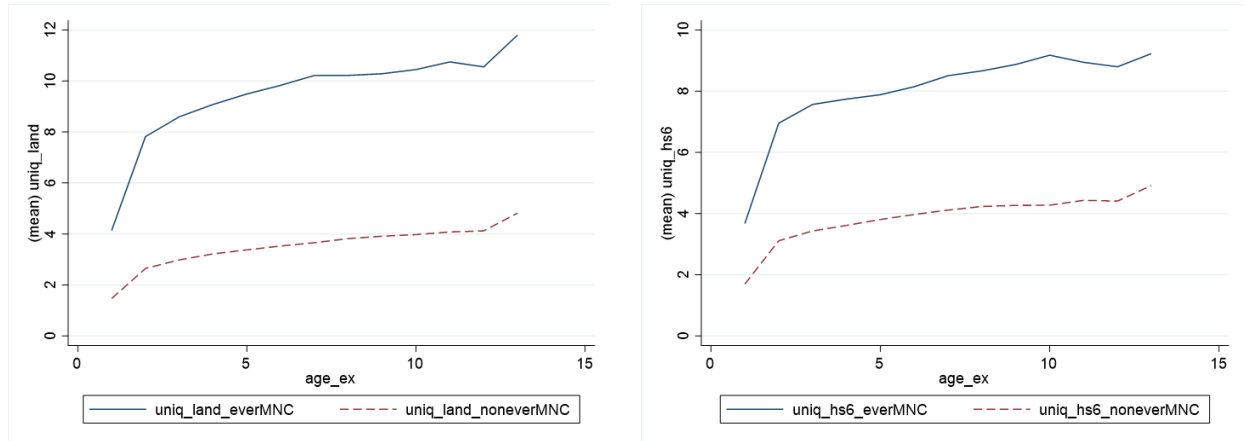
Notes: The sample covers all firms exported in 1994–2006 period and includes firms that exported in 1993. The left part of the table reports annual average firm characteristics for all exporters in the sample. The right part reports firm characteristics of new exporters, where a new exporter in year t is a firm that did not export in year $t - 1$; re-entrants are therefore treated as new exporters. An ever-MNC is a firm that, at any point between 1993 and 2006, was owned by a foreign firm or had foreign affiliates. Among new exporters, a Born Global firm is one created at most one year before starting exports, while a Non Born Global firm is one created two or more years before exporting. Exports, sales, and capital are expressed in thousands of euros. Export intensity is defined as exports divided by total sales.

C Appendix: MNC

C.1 Descriptive Statistics

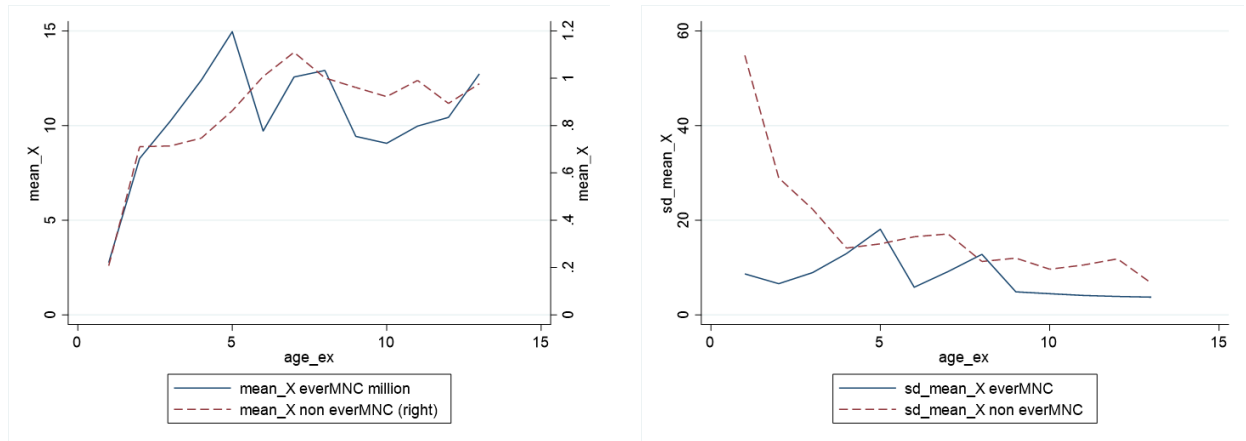
In this subsection we present the same descriptive statistics as in Section 3.1, but on a different sample. In the main text we restricted the analysis to firms whose export spell lasted five consecutive years or more. This was to disentangle pure age patterns from composition effects due to different spell lengths. In what follows we show the same descriptive statistics for all exporters at all ages, independent of attrition.

Tab_A.1: Number of exports destinations (left) and export products (right), by age and multinational status



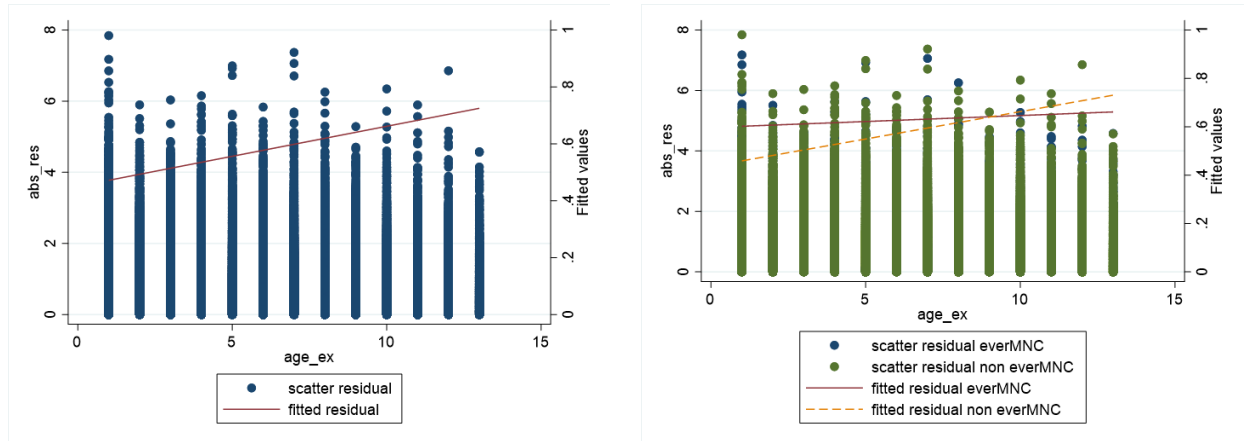
Notes: The sample covers the period from 1994 to 2006. Destinations are defined at the country level; products are defined at the HS6 level. The plot for MNCs new exporters is represented by solid blue curve, and the plot for non-MNCs new exporters is represented by dashed red curve.

Tab_A.2: Export value by age and multinational status: Cross-sectional mean (left panel) and normalized standard deviation (right panel).



Notes: The sample covers the period from 1994 to 2006. Export values are expressed in millions of euros at the firm level. The normalized standard deviation is defined as the ratio of the standard deviation to the mean. The standard deviation is computed across firms by exporting age and multinational status. The plot for MNCs new exporters is represented by solid blue curve, and the plot for non-MNCs new exporters is represented by dashed red curve.

Tab_A.3: Residuals of a regression of export value on age (in absolute value) and fitted line: all firms (left panel) and everMNCs/non everMNCs (right panel)



Notes: The sample covers the period from 1994 to 2006. The residuals are plotted in absolute value (left axis), based on a regression of the logarithm of export value on exporting age at the firm level, with firm-specific slopes and firm fixed effects. The right axis shows the fitted line values. In the right panel, where we distinguish between MNC and non-MNC residuals, the fitted line for MNCs is shown as a solid red line and the fitted line for non-MNCs is shown as a dashed orange line.

C.2 Regression Results

C.2.1 Robustness check: ever-MNCs only

Table 4 shows the growth results using a sample consisting only of ever MNCs. Ever MNCs experience higher growth in the first year-market-product.

Tab_A.4: Growth Results for Ever MNCs

	(1)	(2)
	Growth	Growth
α_{YMP}	0.177 (0.045)***	0.204 (0.041)***
gravity	yes	yes
firm level	yes	yes
year FE	yes	yes
country FE	yes	yes
product FE	yes	
firm FE		yes
N	282568	282568
R^2	0.033	0.044

Notes: The sample covers the period from 1994 to 2006 and includes only Ever MNCs. We use a linear regression. Growth rates are computed from age 2. In specifications (1), we use product fixed effects, and in specifications (2), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * p<0.10, ** p<.05, *** p<.01

Table 5 presents the exit results using a sample consisting solely of exporters that have ever been MNCs. Column 1 includes product fixed effects, while Column 2 includes firm fixed effects. The coefficient β_{YMP} is insignificant for both specifications. Ever MNCs do not exit more or less in the first-year-first-market-first-product compared to other year-market-product triplets.

Tab_A.5: Exit Results for Ever MNCs

	(1)	(2)
	Exit	Exit
β_{YMP}	0.024 (0.023)	-0.022 (0.018)
gravity	yes	yes
firm level	yes	yes
year FE	yes	yes
country FE	yes	yes
product FE	yes	
firm FE		yes
N	437032	436423
R^2	0.226	0.270

Notes: The sample covers the period from 1994 to 2006 and includes only Ever MNCs. We use a linear regression. Year 2006 is used to identify the exit in 2005. In specifications (1), we use product fixed effects, and in specifications (2), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * $p < 0.10$, ** $p < .05$, *** $p < .01$

Table 6 shows the entry results using a sample with only ever MNC. The results are consistent with those in the main text.

Tab_A.6: Entry Regressions of
ever MNC

	(1)	(2)
	Entry	Entry
γ_Y	0.00035 (0.00025)	-0.00054 (0.00028)*
γ_{YM}	0.01729 (0.00197)***	0.01713 (0.00181)***
γ_{YP}	0.00287 (0.00067)***	0.00305 (0.00060)***
gravity	yes	yes
firm level	yes	yes
year FE	yes	yes
country FE	yes	yes
product FE	yes	
firm FE		yes
N	11384410	11384410
R^2	0.01142	0.01254

Notes: The sample covers the period from 1994 to 2006 and includes only Ever MNCs. We use a linear regression. In specifications (1), we use product fixed effects, and in specifications (2), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C.2.2 Robustness check: exit regressions allowing for FDI

In this robustness check, we use an alternative definition of exit in which firms that conduct FDI (i.e., have foreign affiliates) in a given destination country are considered as continuing to serve that market, even if their recorded exports drop to zero.

We use information on French firms' foreign affiliates from the LIFI dataset, which reports, for each year, whether a French firm has an affiliate in a specific foreign country. From this, we construct a foreign-affiliate dataset. In our exit dataset, we first keep all firm–country–product–year observations where the original exit dummy equals 1. For each *firm-country* pair that also appears in the foreign-affiliate dataset, we create all pairwise combinations using 'joinby' in Stata. Thus, if a firm has a foreign affiliate in a given country in a given year, we treat the previously “exiting” products as being supplied to that country in that year. We then append this augmented dataset to the original export dataset and drop duplicate observations, since some firms already had affiliates in certain countries and exported simultaneously prior to exiting a country-product pair.³³

³³Because we use a definition of permanent exit, we add observations for firms that had an affiliate in a given

This procedure allows us to construct an alternative exit measure that excludes cases where a firm remains present in the destination through FDI after recorded exports of a given product cease. Formally, for all observations in the baseline exit dataset, we have:

$$\text{Exit}'_{ijpt} = \begin{cases} 0 & \text{if } \text{Exit}_{ijpt} = 1 \text{ and } \text{MNC}_{ij,t+i} = 1 \text{ for some } i \geq 0 \\ \text{Exit}_{ijpt} & \text{otherwise.} \end{cases}$$

The results in Table 7 closely resemble those in the main text, with some coefficients becoming more significant. The findings regarding exit are robust.

Tab.A.7: MNC Exit Regressions Baseline (Exit corrected by FA)

	(1) Exit	(2) Exit	(3) Exit	(4) Exit	(5) Exit	(6) Exit
β_{YMP}	0.141 (0.008)***	-0.008 (0.006)	0.157 (0.008)***	-0.005 (0.006)	0.157 (0.008)***	-0.005 (0.006)
MNC_it	-0.014 (0.006)**	-0.015 (0.011)	-0.017 (0.008)**	-0.017 (0.013)		
β_{YMPmnc}			-0.144 (0.033)***	-0.033 (0.023)		
β_{YMPfo}					-0.138 (0.035)***	-0.025 (0.024)
β_{YMPpa}					-0.120 (0.058)**	-0.101 (0.037)***
FO_it					-0.006 (0.009)	-0.023 (0.013)*
Parent_it					-0.057 (0.024)**	0.011 (0.025)
gravity	yes	yes	yes	yes	yes	yes
firm level	yes	yes	yes	yes	yes	yes
year FE	yes	yes	yes	yes	yes	yes
country FE	yes	yes	yes	yes	yes	yes
product FE	yes		yes		yes	
firm FE		yes		yes		yes
Observations	1716673	1692867	1716673	1692867	1716673	1692867
R^2	0.268	0.342	0.268	0.342	0.269	0.342
$\beta_{YMP} + \beta_{YMPmnc}$			0.014	-0.037*		
$\beta_{YMP} + \beta_{YMPfo}$					0.019	-0.029
$\beta_{YMP} + \beta_{YMPpa}$					0.037	-0.105***

Notes: The sample covers the period from 1994 to 2006. The exit variable is corrected using information on firms' foreign affiliates. We use a linear regression. Year 2006 is used to identify the exit in 2005. In specifications (1), (3), and (5), we use product fixed effects, and in specifications (2), (4), and (6), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * p<0.10, ** p<.05, *** p<.01

Table 8 confirms these results on the subsample of ever MNCs only. Ever MNCs do not exit country and year but did not export simultaneously prior to exiting a country-product pair.

more in the first-year-first-market-first-product compared to other year-market-product triplets.

Tab.A.8: Exit Regressions of ever MNC (Exit corrected by FA)

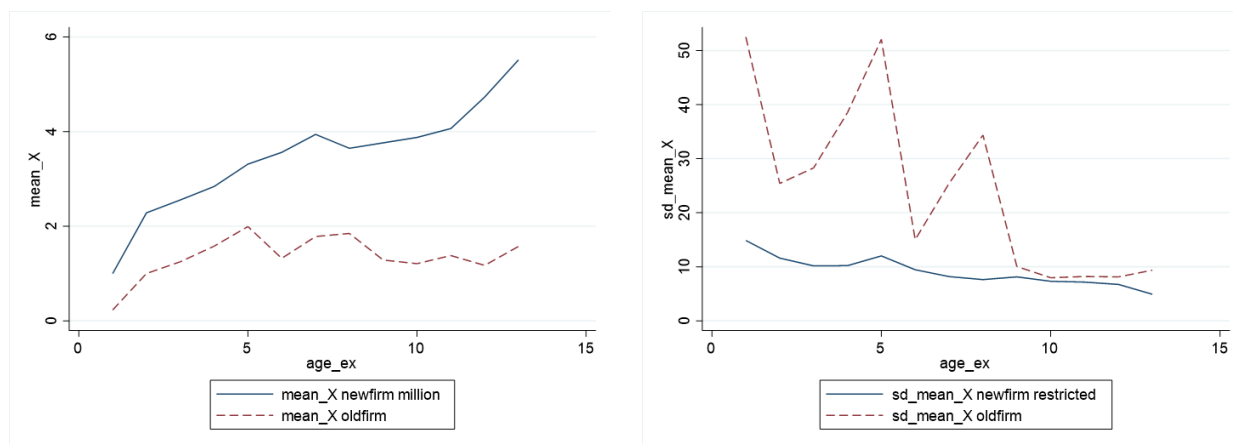
	(1)	(2)
	Exit	Exit
β_{YMP}	0.030 (0.023)	-0.013 (0.017)
gravity	yes	yes
firm level	yes	yes
year FE	yes	yes
country FE	yes	yes
product FE	yes	
firm FE		yes
N	510181	509515
R^2	0.242	0.288

Notes: The sample covers the period from 1994 to 2006 and includes only Ever MNCs. The exit variable is corrected using information on firms' foreign affiliates. We use a linear regression. Year 2006 is used to identify the exit in 2005. In specifications (1), we use product fixed effects, and in specifications (2), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * p<0.10, ** p<.05, *** p<.01

D Appendix: New Firms

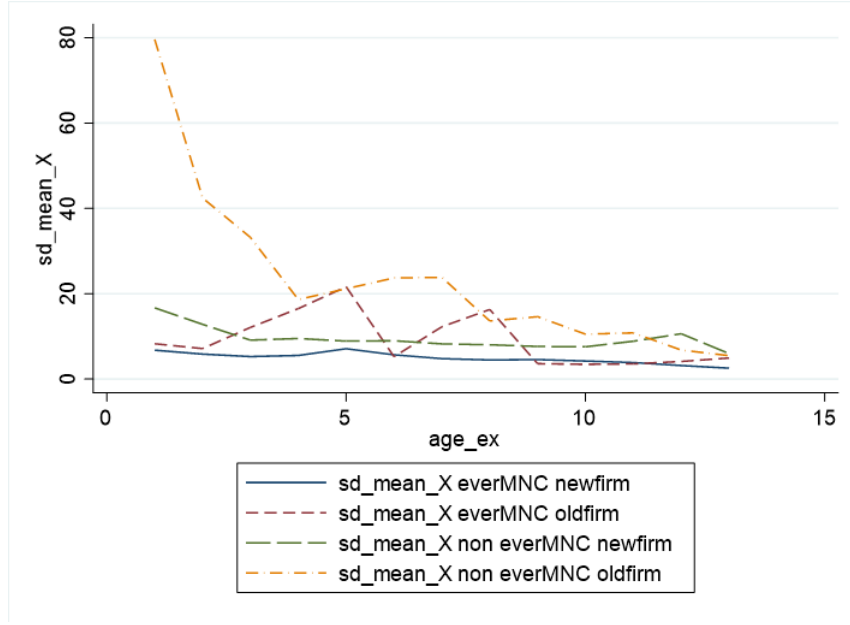
D.1 Descriptive Statistics

Tab_A.4: Export value by age and BG status: Cross-sectional mean (left panel) and normalized standard deviation (right panel)



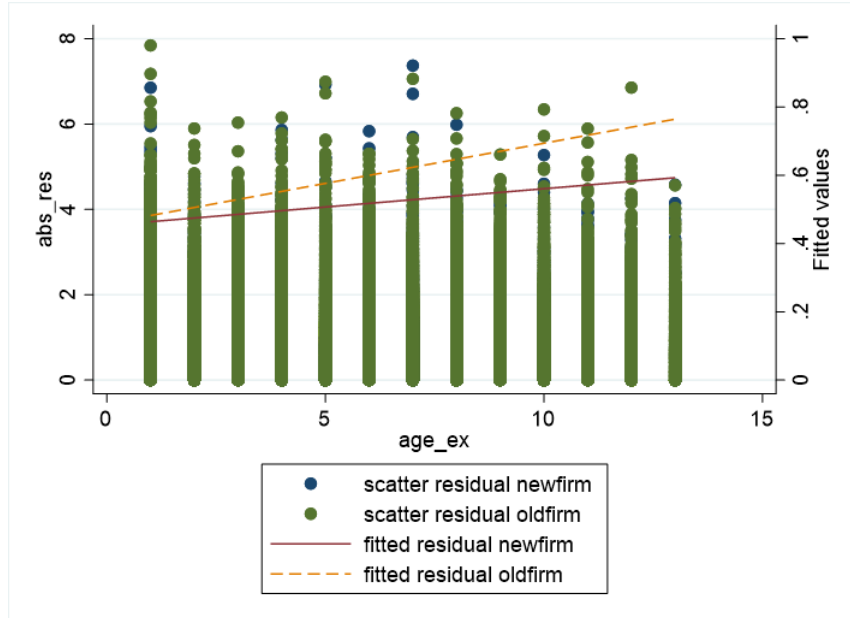
Notes: The sample covers the period from 1994 to 2006. Export values are expressed in millions of euros at the firm level. The normalized standard deviation is defined as the ratio of the standard deviation to the mean. The standard deviation is computed across firms by exporting age and BG status. The plot for BG new exporters is represented by solid blue curve, and the plot for non-BG new exporters is represented by dashed red curve.

Tab_A.5: MNCs $\times$ BGs normalized standard deviation



Notes: The sample covers the period from 1994 to 2006. Export values are expressed in millions of euros at the firm level. The normalized standard deviation is defined as the ratio of the standard deviation to the mean. The standard deviation is computed across firms by exporting age, MNC status, and BG status. Among new exporters, the plots for existing non-MNC firms, new non-MNC firms, existing MNCs, and new MNCs are represented by the orange dot-dashed line, the green long-dashed line, the red short-dashed line, and the solid blue line, respectively.

Tab_A.6: Residuals and fitted line new/old firm



Notes: The sample covers the period from 1994 to 2006. The residuals are plotted in absolute value (left axis), based on a regression of the logarithm of export value on exporting age at the firm level, with firm-specific slopes and firm fixed effects. The right axis shows the fitted line values. We distinguish between BGs and non-BGs residuals, the fitted line for BGs is shown as a solid red line and the fitted line for non-BGs is shown as a dashed yellow line.

D.2 Regression Results

D.2.1 Robustness check: BGs only

Table 9 shows the growth results using a sample consisting only of BGs. BGs experience higher growth in the first year-market-product as in the main specification.

Tab.A.9: Growth Results for only BGs

	(1)	(2)
	Growth	Growth
α_{YMP}	0.181 (0.035)***	0.202 (0.034)***
gravity	yes	yes
firm level	yes	yes
year FE	yes	yes
country FE	yes	yes
product FE	yes	
firm FE		yes
N	416505	416505
R^2	0.026	0.049

Notes: The sample covers the period from 1994 to 2006 and include only observations of BGs. We use a linear regression. Growth rates are computed from age 2. In specifications (1), we use product fixed effects, and in specifications (2), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * $p < 0.10$, ** $p < .05$, *** $p < .01$

Table 10 presents the exit results using a sample consisting solely of exporters that are BGs. Column 1 includes product fixed effects, while Column 2 includes firm fixed effects. As in the main specification, the coefficient β_{YMP} is significant using product fixed effect and is not significant using firm fixed effect. As our preferred specification is the one using product fixed effect. BGs exit more in the first-year-first-market-first-product compared to other year-market-product triplets.

Tab_A.10: Exit Results for only BGs

	(1)	(2)
	Exit	Exit
β_{YMP}	0.072 (0.021)***	-0.013 (0.018)
gravity	yes	yes
firm level	yes	yes
year FE	yes	yes
country FE	yes	yes
product FE	yes	
firm FE		yes
N	673830	672285
R^2	0.221	0.288

Notes: The sample covers the period from 1994 to 2006 and include only observations of BGs. We use a linear regression. Year 2006 is used to identify the exit in 2005. In specifications (1), we use product fixed effects, and in specifications (2), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * p<0.10, ** p<.05, *** p<.01

Table 11 shows the entry results using a sample with only BG observations. The results are consistent with those in the main text.

Tab_A.11: Entry Regressions of only BGs

	(1)	(2)
	Entry	Entry
γ_Y	0.00050 (0.00016)***	-0.00026 (0.00017)
γ_{YM}	0.02184 (0.00187)***	0.02226 (0.00177)***
γ_{YP}	0.00193 (0.00035)***	0.00192 (0.00034)***
gravity	yes	yes
firm level	yes	yes
year FE	yes	yes
country FE	yes	yes
product FE	yes	
firm FE		yes
N	21291347	21291347
R^2	0.01190	0.01360

Notes: The sample covers the period from 1994 to 2006 and include only observations of BGs. We use a linear regression. In specifications (1), we use product fixed effects, and in specifications (2), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * $p < 0.10$, ** $p < .05$, *** $p < .01$

D.2.2 Robustness check: exit regressions allowing for FDI

In this robustness check, we use an alternative definition of exit in which firms that conduct FDI (i.e., have foreign affiliates) in a given destination country are considered as continuing to serve that market, even if their recorded exports drop to zero. For the detailed steps used to construct this alternative exit measure, see Section C.2.2 in Appendix.

The results in Table 12 closely resemble those in the main text. The findings regarding exit are robust.

Tab.A.12: BGs Exit Regressions Baseline (Exit corrected by FA)

	(1)	(2)	(3)	(4)
	Exit	Exit	Exit	Exit
β_{YMP}	0.137 (0.008)***	-0.008 (0.006)	0.131 (0.010)***	-0.022 (0.007)***
BG	-0.011 (0.004)***	0.000 (.)	-0.001 (0.006)	0.000 (.)
β_{YMPbg}			-0.063 (0.021)***	0.021 (0.018)
gravity	yes	yes	yes	yes
firm level	yes	yes	yes	yes
year FE	yes	yes	yes	yes
country FE	yes	yes	yes	yes
product FE	yes		yes	
firm FE		yes		yes
N	1709493	1685802	1709493	1685802
R^2	0.268	0.341	0.268	0.342
$\beta_{YMP} + \beta_{YMPbg}$			0.068***	-0.001

Notes: The sample covers the period from 1994 to 2006. The exit variable is corrected using information on firms' foreign affiliates. We use a linear regression. Year 2006 is used to identify the exit in 2005. In specifications (1), and (3), we use product fixed effects, and in specifications (2), and (4), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * p<0.10, ** p<.05, *** p<.01

Table 13 confirms these results on the subsample of BG only.

Tab.A.13: Exit Regressions of only BGs (Exit corrected by FA)

	(1)	(2)
	Exit	Exit
β_{YMP}	0.076 (0.019)***	0.003 (0.017)
gravity	yes	yes
firm level	yes	yes
year FE	yes	yes
country FE	yes	yes
product FE	yes	
firm FE		yes
N	789206	787599
R^2	0.256	0.319

Notes: The sample covers the period from 1994 to 2006 and includes only observations of BGs. The exit variable is corrected using information on firms' foreign affiliates. We use a linear regression. Year 2006 is used to identify the exit in 2005. In specifications (1), we use product fixed effects, and in specifications (2), we use firm fixed effects. See Appendix A.2.5 for the list and definitions of the gravity and firm-level variables used. * p<0.10, ** p<.05, *** p<.01